Statistical Inference for Linear Stochastic Approximation with Richardson-Romberg Extrapolation

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Abstract
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1. Introduction

Stochastic approximation (SA) algorithms are a cornerstone of modern computational statistics, optimization, and reinforcement learning. Introduced by Robbins and Monro (1951), these iterative procedures provide a principled way to find roots of equations or optimize objectives when only noisy observations are available. A particularly important subclass is the *linear stochastic approximation* (LSA) algorithm, which arises naturally in temporal-difference (TD) learning, policy evaluation, and stochastic gradient descent for linear models.

In this work, we study the LSA recursion with a *constant step size* $\alpha > 0$:

$$\theta_k^{(\alpha)} = \theta_{k-1}^{(\alpha)} - \alpha\Big\{A(Z_k)\theta_{k-1}^{(\alpha)} - b(Z_k)\Big\}, \quad k \geq 1, \tag{1}$$

where $\{Z_k\}_{k\in\mathbb{N}}$ is a time-homogeneous Markov chain on a measurable space $(Z, \mathcal{Z})$ with transition kernel Q and unique invariant distribution π. The mappings $A : Z \to \mathbb{R}^{d\times d}$ and $b : Z \to \mathbb{R}^d$ are measurable functions satisfying $\overline{A} \coloneqq \int_Z A(z)d\pi(z)$ and $\overline{b} \coloneqq \int_Z b(z)d\pi(z)$. Assuming that $-\overline{A}$ is a *Hurwitz matrix* (all eigenvalues have strictly negative real parts), the target parameter $\theta^* = \overline{A}^{-1}\overline{b}$ is uniquely defined.

The *Polyak–Ruppert averaging* procedure (Polyak, 1990; Ruppert, 1988) provides an effective variance reduction technique. Given a burn-in period $n_0 \geq 0$, the averaged iterate is defined as

$$\overline{\theta}_n^{(\alpha)} = \frac{1}{n - n_0}\sum_{k=n_0}^{n-1} \theta_k^{(\alpha)}. \tag{2}$$

1.1. Bias of constant step-size iterates

The use of a constant step size $\alpha > 0$ offers several practical advantages: it enables geometrically fast forgetting of the initial condition (Dieuleveut, Durmus, and Bach, 2020) and simplifies hyperparameter tuning compared to diminishing step-size schedules. However, unlike the classical regime $\alpha_k \to 0$ with $\sum \alpha_k = \infty$ and $\sum \alpha_k^2 < \infty$, a constant step size produces iterates that converge only *in distribution* to a stationary measure Π_α, rather than almost surely to θ^*. The stationary expectation $\mathbb{E}\Big[\theta_\infty^{(\alpha)}\Big]$ is generally *biased* with respect to θ^*, and this bias cannot be eliminated by Polyak–Ruppert averaging alone.

As shown in Levin, Naumov, and Samsonov (2025), the stationary bias has a leading linear term in α:

$$\lim_{n\to\infty} \mathbb{E}\Big[\theta_n^{(\alpha)}\Big] = \theta^* + \alpha\Delta + O\big(\alpha^{3/2}\big), \tag{3}$$

where $\Delta = \overline{A}^{-1} \sum_{k=1}^{\infty} \mathbb{E}\left[\left\{Q^k \tilde{A}(Z_\infty)\right\}\epsilon(Z_\infty)\right]$ depends on the correlation structure of the Markov chain, and $\tilde{A}(z) = A(z) - \overline{A}$ is the centered matrix-valued function. Under stronger expansion assumptions, the power-series approach of Huo, Chen, and Xie (2023) gives higher-order bias expansions in integer powers of α; in the Levin decomposition, the first misadjustment bias component itself has an $O(\alpha^2)$ remainder after the leading $\alpha\Delta$ term.

1.2. Richardson–Romberg extrapolation

To eliminate the leading $O(\alpha)$ bias term, we employ the *Richardson–Romberg* (RR) *extrapolation* procedure. Two LSA sequences are run *on the same Markov chain trajectory* $\{Z_k\}$ with step sizes α and 2α, and the RR iterate is formed as

$$\overline{\theta}_n^{(\alpha,\,\mathrm{RR})} = 2\overline{\theta}_n^{(\alpha)} - \overline{\theta}_n^{(2\alpha)}. \tag{4}$$

Since both sequences share the same noise realization, the leading bias term $\alpha\Delta$ cancels, leaving a residual bias of order $O(\alpha^{3/2})$ or higher (Levin et al., 2025).

More generally, one can consider the multi-level extrapolation with M step sizes $\mathcal{A} = \{\alpha_1, ..., \alpha_M\}$ and coefficients $\{h_m\}$ determined by the Vandermonde system (Huo et al., 2023):

$$\sum_{m=1}^{M} h_m = 1, \quad \sum_{m=1}^{M} h_m \alpha_m^l = 0, \quad l = 1, ..., M-1, \tag{5}$$

which cancels successive powers in settings where such a power-series expansion is available.

1.3. Problem statement and goals

The high-order moment bounds for the RR iterate $\overline{\theta}_n^{(\alpha,\,\mathrm{RR})}$ established in Levin et al. (2025) show that the leading error term scales as $\sqrt{\mathrm{Tr}\ \Sigma_\epsilon^{(\mathrm{M})}} \cdot n^{-1/2}$, where $\Sigma_\epsilon^{(\mathrm{M})}$ is the Markovian noise covariance defined below in the key quantities subsection, matching the minimax optimal rate. Berry–Esseen type bounds and bootstrap inference procedures for the *standard* Polyak–Ruppert average $\overline{\theta}_n$ (without extrapolation) under Markovian noise have been obtained in Samsonov, Sheshukova, Moulines, and Naumov (2025).

However, the *distributional approximation* — specifically, the central limit theorem and Berry–Esseen bounds — for the *averaged Richardson–Romberg iterates* $\overline{\theta}_n^{(\alpha,\,\mathrm{RR})}$ remains an open problem. The main goal of this work is to fill this gap by establishing:

1. A *central limit theorem* for $\sqrt{n}\left(\overline{\theta}_n^{(\alpha,\,\mathrm{RR})} - \theta^*\right)$, identifying the limiting covariance matrix.
2. *Non-asymptotic Berry–Esseen bounds* for the rate of convergence to the Gaussian limit.
3. A practical *inference procedure* (confidence intervals) based on the RR-extrapolated iterates.

1.4. Setting and assumptions

We now formalize the setting and state the assumptions that will be used throughout this work.

Let $\{Z_k\}_{k\in\mathbb{N}}$ be a Markov chain on a complete separable metric space $(Z, \mathcal{Z})$ with transition kernel Q.
Assumption 1 (Uniform geometric ergodicity). The kernel Q admits a unique invariant distribution π and is *uniformly geometrically ergodic*: there exists $t_{\mathrm{mix}} \in \mathbb{N}^*$ such that for all $k \in \mathbb{N}^*$,

$$\Delta\left(Q^k\right) := \sup_{z,z' \in Z} \frac{1}{2} \left\|Q^k(z,\cdot) - Q^k(z',\cdot)\right\|_{\mathrm{TV}} \leq (1/4)^{\lfloor k/t_{\mathrm{mix}} \rfloor}. \tag{6}$$

Equivalently, there exist constants $\zeta > 0$ and $\rho \in (0,1)$ such that $\sup_z \|Q^k(z,\cdot) - \pi\|_{\mathrm{TV}} \leq \zeta\rho^k$ for all $k \geq 1$.
Assumption 2 (Hurwitz condition and boundedness). The matrix $-\overline{A}$ is Hurwitz, i.e., all eigenvalues of $\overline{A}$ have strictly positive real parts. Moreover,

$$C_A := \max\left(\sup_{z\in Z}\|A(z)\|, \sup_{z\in Z}\|\tilde{A}(z)\|\right) < \infty, \tag{7}$$

where $\tilde{A}(z) := A(z) - \overline{A}$.

Assumption 3 (Noise regularity). The noise function $\epsilon(z) = \tilde{A}(z)\theta^* - \tilde{b}(z)$, where $\tilde{b}(z) = b(z) - \overline{b}$, satisfies

$$\|\epsilon\|_\infty := \sup_{z\in Z}\|\epsilon(z)\| < +\infty. \tag{8}$$

Under Assumptions 1–3, the error $\theta_k^{(\alpha)} - \theta^*$ satisfies the recursion

$$\theta_k^{(\alpha)} - \theta^* = (I - \alpha A(Z_k))\left(\theta_{k-1}^{(\alpha)} - \theta^*\right) - \alpha\epsilon(Z_k). \tag{9}$$

1.5. Key quantities

The *Markovian noise covariance matrix* captures both the marginal variance and the temporal correlations of the noise:

$$\Sigma_\epsilon^{(\mathrm{M})} = \mathbb{E}_\pi\left[\epsilon(Z_0)\epsilon(Z_0)^\top\right] + \sum_{\ell=1}^{\infty}\left(\mathbb{E}_\pi\left[\epsilon(Z_0)\epsilon(Z_\ell)^\top\right] + \mathbb{E}_\pi\left[\epsilon(Z_\ell)\epsilon(Z_0)^\top\right]\right). \tag{10}$$

This matrix is the limiting covariance in the Markov chain CLT for the partial sums $n^{-1/2}\sum_{t=0}^{n-1}\epsilon(Z_t)$ (cf. Douc et al., 2018, Theorem 21.2.10).

The *asymptotically optimal covariance matrix* is given by

$$\Sigma_\infty = \overline{A}^{-1}\Sigma_\epsilon^{(\mathrm{M})}\left(\overline{A}^{-1}\right)^\top. \tag{11}$$

This is the covariance target attained by the averaged linearized recursion. We call it optimal in the usual averaged-SA sense; a full Hájek–Le Cam optimality statement would require an additional local-asymptotic experiment argument, which is not part of this thesis.

The *Lyapunov equation* plays a central role in the contraction analysis. For any $P = P^\top \succ 0$, there exists a unique $Q = Q^\top \succ 0$ satisfying $\overline{A}^\top Q + Q\overline{A} = P$. Defining $a = \lambda_{\min}(P)/(2\,\|Q\|)$ and $\kappa_Q = \lambda_{\max}(Q)/\lambda_{\min}(Q)$, the key contraction property holds: for all $\alpha \in [0, \alpha_\infty]$,

$$\|I - \alpha\overline{A}\|_Q^2 \le 1 - \alpha a. \tag{12}$$

Since the iterates $\theta_k^{(\alpha)}$ alone are generally not Markovian (due to the Markovian noise), we consider the *joint process* $\left(\theta_k^{(\alpha)}, Z_{k+1}\right)$ with kernel

$$\overline{P}_\alpha f(\theta, z) = \int_Z Q(z, dz')f(F_{z'}(\theta), z'), \tag{13}$$

where $F_z(\theta) = (I - \alpha A(z))\theta + \alpha b(z)$. Under Assumptions 1–3, this joint chain admits a unique invariant distribution Π_α for sufficiently small $\alpha > 0$ (Levin et al., 2025).

2. Zeroth-Order Richardson–Romberg Difference

2.1. LSA Error Decomposition

We consider the recursion

$$\theta_k = \theta_{k-1} - \alpha_k (A(Z_k)\theta_{k-1} - b(Z_k)), \quad \alpha_k = \alpha = \text{const.} \tag{14}$$

Define the transition products

$$\Gamma_{m:k} = \prod_{l=m}^{k} (I - \alpha A(Z_l)). \tag{15}$$

Write $B_\alpha \coloneqq I - \alpha \overline{A}$ and introduce the deterministic-product linearized term

$$J_k^{(0,\alpha)} = -\alpha \sum_{l=1}^{k} B_\alpha^{k-l}\, \epsilon(Z_l), \quad J_0^{(0,\alpha)} = 0. \tag{16}$$

The difference between the exact random-product expansion and this deterministic-product term is denoted by $R_k^{(\alpha)}$:

$$\theta_k^{(\alpha)} - \theta^* = J_k^{(0,\alpha)} + B_\alpha^k (\theta_0 - \theta^*) + R_k^{(\alpha)}. \tag{17}$$

This convention matches the weight decomposition in Section 4.1.

A standard PR-averaged decomposition (cf. Chapter 4 for the full derivation) yields

$$\sqrt{n}\left(\overline{\theta}_n^{(\alpha)} - \theta^*\right) = W + D_1, \tag{18}$$

where the leading martingale-like term is

$$W = -\frac{1}{\sqrt{n}} \sum_{l=1}^{n-1} Q_l\, \epsilon(Z_l), \quad Q_l = \alpha \sum_{k=l}^{n-1} B_\alpha^{k-l}, \tag{19}$$

and the residual term is

$$D_1 = \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} B_\alpha^k (\theta_0 - \theta^*) + \frac{1}{\sqrt{n}} \sum_{k=1}^{n-1} R_k^{(\alpha)}. \tag{20}$$

2.2. RR Combination and the $\tilde{J}_n^{(0,\alpha)}$ Term

Applying the deterministic-product decomposition of the previous subsection separately to step sizes α and 2α, and writing $B_{2\alpha} \coloneqq I - 2\alpha\overline{A}$, the Richardson–Romberg combination has the form

$$\begin{aligned} \theta_n^{(\mathrm{RR},\alpha)} - \theta^* &= [2B_\alpha^n - B_{2\alpha}^n](\theta_0 - \theta^*) \\ &\quad + \left[2J_n^{(0,\alpha)} - J_n^{(0,2\alpha)}\right] + \left[2R_n^{(\alpha)} - R_n^{(2\alpha)}\right]. \end{aligned} \tag{21}$$

The first bracket is the deterministic transient in this convention. The second bracket is the linearized stochastic RR difference, and the last one is the higher-order random-product remainder. In this subsection we focus on the zeroth-order RR difference, which we denote

$$\tilde{J}_n^{(0,\alpha)} = 2J_n^{(0,\alpha)} - J_n^{(0,2\alpha)}, \tag{22}$$

where, by definition,

$$J_n^{(0,\alpha)} = -\alpha \sum_{j=1}^{n} \left(I - \alpha\overline{A}\right)^{n-j} \epsilon(Z_j). \tag{23}$$

Substituting and using the elementary identity $X^m - Y^m = (X - Y)\sum_{i=1}^{m} X^{i-1}Y^{m-i}$ with $X = I - \alpha\overline{A}$, $Y = I - 2\alpha\overline{A}$, $X - Y = \alpha\overline{A}$, we get

$$\begin{aligned}
\tilde{J}_n^{(0,\alpha)} &= -2\alpha \sum_{j=1}^{n} \Big[\big(I - \alpha\overline{A}\big)^{n-j} - \big(I - 2\alpha\overline{A}\big)^{n-j} \Big] \epsilon\big(Z_j\big) \\
&= -2\alpha^2 \overline{A} \sum_{j=1}^{n} \underbrace{\sum_{i=1}^{n-j} \big(I - \alpha\overline{A}\big)^{i-1} \big(I - 2\alpha\overline{A}\big)^{n-j-i} \epsilon\big(Z_j\big)}_{=:H_j^{(n)}} \\
&= -2\alpha^2 \overline{A} \sum_{j=1}^{n-1} H_j^{(n)}\, \epsilon\big(Z_j\big),
\end{aligned} \tag{24}$$

where the last equality uses $H_n^{(n)} = 0$ because the inner sum is empty. The extra factor $\alpha\overline{A}$ pulled out front is the source of the additional α-decay of the RR difference compared to a single LSA trajectory.

2.3. Norm Estimate for $H_j^{(n)}$

To bound $\|H_j^{(n)}\|$ we use the following standard result.

Lemma 1. Let $-\overline{A}$ be a Hurwitz matrix. Then for any $P = P^\top \succ 0$ there exists a unique $Q = Q^\top \succ 0$ satisfying

$$\overline{A}^\top Q + Q\overline{A} = P. \tag{25}$$

Moreover, letting

$$a := \frac{\lambda_{\min}(P)}{2\,\|Q\|}, \quad \alpha_\infty := \min\left(\frac{\lambda_{\min}(P)}{2\kappa_Q\, \|\overline{A}\|_Q^2}, \frac{\|Q\|}{\lambda_{\min}(P)} \right), \tag{26}$$

with $\kappa_Q := \lambda_{\max}(Q)/\lambda_{\min}(Q)$, one has for all $\alpha \in [0, \alpha_\infty]$:

$$\alpha a \le 1/2, \quad \|I - \alpha\overline{A}\|_Q^2 \le 1 - \alpha a. \tag{27}$$

For $1 \le j \le n-1$, we estimate $\|H_j^{(n)}\|$ by combining the triangle inequality, submultiplicativity, the equivalence $\| X \| \le \kappa_Q^{1/2} \| X \|_Q$ (applied to **each** operator-norm factor, hence two factors of $\kappa_Q^{1/2}$ multiplying to κ_Q), and the Lyapunov contraction at step sizes α and 2α:

$$\begin{aligned}
\|H_j^{(n)}\| &\le \sum_{i=1}^{n-j} \|I - \alpha\overline{A}\|^{i-1}\, \|I - 2\alpha\overline{A}\|^{n-j-i} && \text{(triangle + submult.)} \\
&\le \kappa_Q \sum_{i=1}^{n-j} (1 - \alpha a)^{(i-1)/2} (1 - 2\alpha a)^{(n-j-i)/2} && \text{(equiv. + Lyapunov)} \\
&= \kappa_Q (1 - \alpha a)^{(n-j-1)/2} \sum_{k=0}^{n-j-1} \left(\frac{1 - 2\alpha a}{1 - \alpha a} \right)^{k/2} && \text{(reindex } k = n - j - i\text{)} \\
&\le \kappa_Q (1 - \alpha a)^{(n-j-1)/2} \frac{1}{1 - \sqrt{\frac{1-2\alpha a}{1-\alpha a}}} && \text{(geometric series).}
\end{aligned} \tag{28}$$

For $\alpha a \le 1/2$ the geometric rate is bounded below by an **elementary** inequality: combining $\sqrt{(1 - 2\alpha a)/(1 - \alpha a)} \le \sqrt{1 - \alpha a}$ (since $(1 - 2\alpha a) \le (1 - \alpha a)^2$) with $1 - \sqrt{1 - x} \ge x/2$ on $[0, 1]$ gives

$$1 - \sqrt{\frac{1 - 2\alpha a}{1 - \alpha a}} \ge 1 - \sqrt{1 - \alpha a} \ge \frac{\alpha a}{2}, \quad \text{i.e.} \quad \frac{1}{1 - \sqrt{\frac{1-2\alpha a}{1-\alpha a}}} \le \frac{2}{\alpha a}. \tag{29}$$

Defining the chapter-local constant

$$\overline{C}_A \coloneqq \kappa_Q \tag{30}$$

(distinct from the assumption-2 sup-norm constant C_A, which enters separately via $\| \overline{A} \|$ in the next subsection), we arrive at the final estimate

$$\|H_j^{(n)}\| \le \overline{C}_A \, (1-\alpha a)^{(n-j-1)/2} \, \frac{2}{\alpha a}. \tag{31}$$

The kernel decays geometrically in $n-j$ at rate $\sqrt{1-\alpha a}$ but its summed weight is of order $1/(\alpha a)$. The next subsection keeps the remaining powers of a explicit when this kernel is multiplied by the prefactor α^2 in $\tilde{J}_n^{(0,\alpha)}$.

2.4. Scalar L^p Bound for the Zeroth-Order Term

The expression

$$\tilde{J}_n^{(0,\alpha)} = -\sum_{j=1}^{n-1} 2\alpha^2 \overline{A} H_j^{(n)} \, \epsilon\big(Z_j\big) \tag{32}$$

is a weighted sum of values of the centered noise function ϵ along the Markov chain. In the Berry–Esseen argument below only fixed scalar projections are used, so we state the concentration input in scalar form.

Lemma 2. Assume **UGE 1**. Let $\{g_i\}_{i=1}^n$ be a family of measurable functions $g_i : Z \to \mathbb{R}$ such that

$$c_i = \|g_i\|_\infty < \infty \quad \text{for all } i \ge 1, \quad \pi(g_i) = 0 \quad \text{for all } i \in \{1, ..., n\}. \tag{33}$$

Then, for any initial distribution ξ on $(Z, \mathcal{Z})$, any $n \in \mathbb{N}$ and any $t \ge 0$,

$$\mathbb{P}_\xi\left(\left|\sum_{i=1}^n g_i(Z_i)\right| \ge t\right) \le 2\exp\left(-\frac{t^2}{2v_n^2}\right), \tag{34}$$

where

$$v_n = 8\left(\sum_{i=1}^n c_i^2\right)^{1/2} \sqrt{t_{\text{mix}}}. \tag{35}$$

Proof. The proof follows the lines of Durmus et al. (2025, Lemma 9).

Fix a deterministic direction $u \in \mathbb{R}^d$. We apply the lemma with $g_j^{u(z)} = -2\alpha^2 u^\top \overline{A} H_j^{(n)} \epsilon(z)$ for $1 \le j \le n-1$ and $g_n^u = 0$. Each g_j^u is centered under π since $\pi(\epsilon) = 0$, so the centering hypothesis holds. Combining the bound $\| \overline{A} \| \le C_A$ (Assumption 2) with the previous subsection, define

$$\tilde{C}_A \coloneqq C_A \, \overline{C}_A = C_A \, \kappa_Q. \tag{36}$$

Then the per-summand bound is

$$\begin{aligned} \|g_j^u\|_\infty &\le 2\alpha^2 \; \|u\| \, \tilde{C}_A (1-\alpha a)^{(n-j-1)/2} \frac{2}{\alpha a} \, \|\epsilon\|_\infty \\ &= \frac{4\alpha \; \|u\| \, \tilde{C}_A \; \|\epsilon\|_\infty}{a} (1-\alpha a)^{(n-j-1)/2}, \quad 1 \le j \le n-1. \end{aligned} \tag{37}$$

Note the prefactor α^2 collapsing to α/a: the $1/(\alpha a)$ blow-up in $H_j^{(n)}$ is absorbed only by one factor of α. Squaring and summing over j:

$$
\begin{aligned}
\sum_{j=1}^{n} \|g_j^u\|_\infty^2 &= \sum_{j=1}^{n-1} \|g_j^u\|_\infty^2 \\
&\le \frac{16\alpha^2 \ \|u\|^2 \, \tilde{C}_A^2 \ \|\epsilon\|_\infty^2}{a^2} \sum_{j=1}^{n-1} (1-\alpha a)^{n-j-1} \\
&\le \frac{16\alpha^2 \ \|u\|^2 \, \tilde{C}_A^2 \ \|\epsilon\|_\infty^2}{a^2} \, \frac{1}{\alpha a} \\
&= \frac{16\alpha \ \|u\|^2 \, \tilde{C}_A^2 \ \|\epsilon\|_\infty^2}{a^3}.
\end{aligned}
\tag{38}
$$

Plugging this into the variance proxy $v_n^2 = 64 \, t_{\mathrm{mix}} \sum_{j=1}^{n} \|g_j^u\|_\infty^2$ from the lemma and defining

$$
\hat{C}_A \coloneqq 32 \, \tilde{C}_A \, \|\epsilon\|_\infty \, \sqrt{\frac{t_{\mathrm{mix}}}{a^3}}, \tag{39}
$$

gives

$$
v_n^2 \le \|u\|^2 \, \hat{C}_A^2 \, \alpha. \tag{40}
$$

To convert the sub-Gaussian tail into a moment bound, we use the following standard fact.

Lemma 3. Let X be an $\mathbb{R}$-valued random variable satisfying

$$
\mathbb{P}(|X| \ge t) \le 2 \exp\left(-\frac{t^2}{2\sigma^2} \right) \quad \text{for all } t \ge 0, \tag{41}
$$

for some $\sigma^2 > 0$. Then, for any $p \ge 2$, it holds that

$$
\mathbb{E}[\![X|^p]\!] \le 2 \, p^{p/2} \, \sigma^p. \tag{42}
$$

Applying the lemma with $X = u^\top \tilde{J}_n^{(0,\alpha)}$ and $\sigma^2 = v_n^2 \le \|u\|^2 \ \hat{C}_A^2 \alpha$ gives, for any $p \ge 2$,

$$
\mathbb{E}^{1/p} \left[\!\!\left[u^\top \tilde{J}_n^{(0,\alpha)} |^p \right]\!\!\right] \le 2^{1/p} \sqrt{p} \, \|u\| \, \hat{C}_A \, \sqrt{\alpha} \le 2\sqrt{p} \, \|u\| \, \hat{C}_A \, \sqrt{\alpha}, \tag{43}
$$

or equivalently

$$
\frac{1}{\sqrt{\alpha}} \, \|u^\top \tilde{J}_n^{(0,\alpha)}\|_{L_p} \le 2\sqrt{p} \, \|u\| \, \hat{C}_A. \tag{44}
$$

Thus every fixed scalar projection of the zeroth-order RR difference is $O(\sqrt{\alpha})$ in L^p, uniformly in n. A Euclidean-norm bound can be recovered in fixed dimension by applying the scalar estimate to a coordinate basis, but no dimension-free vector concentration is claimed here.

3. Last Iterate Analysis

3.1. Centered Bound for the Shifted First-Order Perturbation

The purpose of this section is to isolate the last-iteration weighted term which appears in the analysis of $J_n^{(1,\alpha)}$, and to record a clean L_p bound for its centered part. The proof uses the deterministic-product perturbation expansion from Samsonov et al. (2025, Proposition 9), specialized to the constant-stepsize setting.

Let

$$B = I - \alpha\overline{A}. \tag{45}$$

Throughout this section we assume $0 < \alpha \le \alpha_\infty$, so that

$$\|B^m\|_Q \le (1-\alpha a)^{m/2}, \quad m \ge 0. \tag{46}$$

All constants denoted by C may depend on C_A, κ_Q, and norm equivalence constants, but not on α, n, p, or t_{mix}.

The deterministic-product components are defined by

$$J_n^{(0,\alpha)} = BJ_{n-1}^{(0,\alpha)} - \alpha\epsilon(Z_n), \quad J_0^{(0,\alpha)} = 0, \tag{47}$$

and

$$J_n^{(1,\alpha)} = BJ_{n-1}^{(1,\alpha)} - \alpha\tilde{A}(Z_n)J_{n-1}^{(0,\alpha)}, \quad J_0^{(1,\alpha)} = 0. \tag{48}$$

Hence

$$J_n^{(0,\alpha)} = -\alpha\sum_{k=1}^{n} B^{n-k}\epsilon(Z_k), \tag{49}$$

and

$$\begin{aligned} J_n^{(1,\alpha)} &= -\alpha\sum_{j=1}^{n} B^{n-j}\tilde{A}(Z_j)J_{j-1}^{(0,\alpha)} \\ &= \alpha^2 \sum_{1\le k<j\le n} B^{n-j}\tilde{A}(Z_j)B^{j-1-k}\epsilon(Z_k) \\ &= \alpha^2\sum_{k=1}^{n-1}\sum_{r=1}^{n-k} B^{n-k-r}\tilde{A}(Z_{k+r})B^{r-1}\epsilon(Z_k). \end{aligned} \tag{50}$$

The sign in the second line is positive because $J_{j-1}^{(0,\alpha)}$ already contains the minus sign.

The last display is for $J_n^{(1,\alpha)}$ itself. The last-iteration estimate below operates on a **shifted** version, obtained by inserting one additional left factor B into the recursion:

$$S_n = \sum_{t=0}^{n-1} B^{n-t}\tilde{A}(Z_{t+1})J_t^{(0,\alpha)}. \tag{51}$$

Reindexing $j = t+1$ rewrites S_n in the same form as $J_n^{(1,\alpha)}$ but with one extra power of B:

$$S_n = \sum_{j=1}^{n} B^{n-j+1}\tilde{A}(Z_j)J_{j-1}^{(0,\alpha)} = -\frac{1}{\alpha}\, B\, J_n^{(1,\alpha)}. \tag{52}$$

Thus the corresponding shifted first-order contribution is

$$T_n^{(1,\alpha)} = -\alpha S_n = B\, J_n^{(1,\alpha)}, \tag{53}$$

i.e. $T_n^{(1,\alpha)}$ is exactly $J_n^{(1,\alpha)}$ pre-multiplied by one additional B. Transferring a bound from $T_n^{(1,\alpha)}$ back to $J_n^{(1,\alpha)}$ therefore requires a local inverse bound on $B^{-1} = \left(I - \alpha \overline{A}\right)^{-1}$; this inverse-bound step is used explicitly when the shifted estimate is applied below.

Lemma 4. Assume the Markov chain is started from stationarity, that is, the law of Z_1 is π, and $\pi\left(\tilde{A}\right) = 0$. For every $p \geq 2$,

$$\|S_n - \mathbb{E}S_n\|_{L_p} \leq C\, \|\epsilon\|_\infty \left(p^{3/2} t_{\text{mix}}^{1/2} \frac{1}{a} + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\frac{\alpha}{a}}\right). \tag{54}$$

Consequently,

$$\|T_n^{(1,\alpha)} - \mathbb{E}T_n^{(1,\alpha)}\|_{L_p} \leq C\alpha\, \|\epsilon\|_\infty \left(p^{3/2} t_{\text{mix}}^{1/2} \frac{1}{a} + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\frac{\alpha}{a}}\right). \tag{55}$$

Proof. Since $J_0^{(0,\alpha)} = 0$, the term $t = 0$ in S_n vanishes. Substituting the explicit formula for $J_t^{(0,\alpha)}$ gives

$$\begin{aligned} S_n &= -\alpha \sum_{t=1}^{n-1} \sum_{k=1}^{t} B^{n-t} \tilde{A}(Z_{t+1}) B^{t-k} \epsilon(Z_k) \\ &= -\alpha \sum_{k=1}^{n-1} H_{k+1}^{(w)} \epsilon(Z_k), \end{aligned} \tag{56}$$

where, after the change of summation index $l = t - k + 1$,

$$H_{k+1}^{(w)} = \sum_{l=1}^{n-k} B^{n-k-l+1} \tilde{A}(Z_{k+l}) B^{l-1}. \tag{57}$$

The kernel $H_{k+1}^{(w)}$ acts on the past noise $\epsilon(Z_k)$ through future states Z_{k+l}, so $H_{k+1}^{(w)} \epsilon(Z_k)$ is a future-weighted bilinear functional of the trajectory.

Define

$$\mu_k^{(w)} = \mathbb{E}_\pi\left[H_{k+1}^{(w)} \epsilon(Z_k)\right]. \tag{58}$$

Then

$$S_n - \mathbb{E}S_n = -\alpha \sum_{k=1}^{n-1} \left\{H_{k+1}^{(w)} \epsilon(Z_k) - \mu_k^{(w)}\right\}. \tag{59}$$

Let $\mathcal{F}_k = \sigma(Z_1, ..., Z_k)$. The Markov property gives

$$\mathbb{E}\left[H_{k+1}^{(w)} \epsilon(Z_k) \mid \mathcal{F}_k\right] = v_k^{(w,\varepsilon)}(Z_k), \tag{60}$$

where

$$v_k^{(w)}(z) = \sum_{l=1}^{n-k} B^{n-k-l+1} \left(Q^l \tilde{A}\right)(z) B^{l-1}, \quad v_k^{(w,\varepsilon)}(z) = v_k^{(w)}(z) \epsilon(z). \tag{61}$$

Here Q denotes the one-step Markov transition kernel of $(Z_k)_{k\geq 1}$, acting on bounded matrix-valued functions by integration against the conditional law:

$$\left(Q\tilde{A}\right)(z) = \int \tilde{A}(u)\, Q(z,du) = \mathbb{E}\left[\tilde{A}(Z_{k+1}) \mid Z_k = z\right], \tag{62}$$

and Q^l is its l-fold iterate, the l-step kernel $Q^l(z,du) = \mathbb{P}(Z_{k+l} \in du \mid Z_k = z)$, so that

$$\left(Q^l\tilde{A}\right)(z) = \int \tilde{A}(u)\, Q^l(z,du) = \mathbb{E}\left[\tilde{A}(Z_{k+l}) \mid Z_k = z\right]. \tag{63}$$

In particular $\left(Q^l\tilde{A}\right)(z) \to \pi\left(\tilde{A}\right) = 0$ at the geometric rate dictated by UGE, which is the only fact about Q^l used below.

Under stationarity, $\mu_k^{(w)} = \pi\left(v_k^{(w,\varepsilon)}\right)$. Therefore

$$S_n - \mathbb{E}S_n = -\alpha(U_M + U_R), \tag{64}$$

with

$$U_M = \sum_{k=1}^{n-1}\left\{H_{k+1}^{(w)}\epsilon(Z_k) - v_k^{(w,\varepsilon)}(Z_k)\right\} \tag{65}$$

and

$$U_R = \sum_{k=1}^{n-1}\left\{v_k^{(w,\varepsilon)}(Z_k) - \pi\left(v_k^{(w,\varepsilon)}\right)\right\}. \tag{66}$$

This decomposition splits the centered statistic into two structurally different parts: U_R is an ordinary centered additive functional of the original Markov chain, while U_M is a future-centered bilinear term — the conditional expectation given $\mathcal{F}_k$ vanishes summand-wise, but the summands are not forward martingale differences. We bound each piece separately.

Step 1: bound on U_R. Because $\pi\left(\tilde{A}\right) = 0$, applying Q^l followed by integration against π replaces $\tilde{A}$ by its l-step propagation away from stationarity. UGE then gives the geometric Dobrushin bound

$$\|\left(Q^l\tilde{A}\right)(z)\| = \left\|\int \tilde{A}(u)\{Q^l(z,du) - \pi(du)\}\right\| \le 2C_A\Delta\left(Q^l\right) \le 2C_A(1/4)^{\lfloor l/t_{\text{mix}}\rfloor}. \tag{67}$$

Inserting this into $v_k^{(w)}$ and factoring out the slow rate $(1-\alpha a)^{(n-k)/2}$ from the two B-powers, the inner sum becomes a fast-decaying l-series:

$$\begin{aligned}
\|v_k^{(w)}\|_\infty &\le C\sum_{l=1}^{n-k}(1-\alpha a)^{(n-k-l+1)/2}\Delta\left(Q^l\right)(1-\alpha a)^{(l-1)/2} && \text{(triangle + Lyapunov)}\\
&\le C(1-\alpha a)^{(n-k)/2}\sum_{l=1}^{\infty}(1/4)^{\lfloor l/t_{\text{mix}}\rfloor} && (\text{extend to } l=\infty)\\
&\le Ct_{\text{mix}}(1-\alpha a)^{(n-k)/2} && \text{(geometric block sum).}
\end{aligned} \tag{68}$$

Multiplying by $\|\epsilon\|_\infty$ gives

$$\|v_k^{(w,\varepsilon)}\|_\infty \le Ct_{\text{mix}}\ \|\epsilon\|_\infty\ (1-\alpha a)^{(n-k)/2}. \tag{69}$$

The functions $v_k^{(w,\varepsilon)}(Z_k) - \pi\left(v_k^{(w,\varepsilon)}\right)$ are centered under π and uniformly bounded by the previous display, so the weighted Markov concentration/Rosenthal bound for centered time-dependent functions yields

$$
\begin{aligned}
\|U_R\|_{L_p} &\le C p^{1/2} t_{\mathrm{mix}}^{1/2} \left(\sum_{k=1}^{n-1} \|v_k^{(w,\varepsilon)}\|_\infty^2 \right)^{1/2} && \text{(Rosenthal)} \\
&\le C p^{1/2} t_{\mathrm{mix}}^{3/2} \ \|\epsilon\|_\infty \ \left(\sum_{k=1}^{n-1} (1-\alpha a)^{n-k} \right)^{1/2} && \text{(plug previous bound)} \\
&\le C p^{1/2} t_{\mathrm{mix}}^{3/2} \ \|\epsilon\|_\infty \ \frac{1}{\sqrt{\alpha a}} && \text{(geometric series).}
\end{aligned} \tag{70}
$$

Step 2: bound on U_M. Project onto a deterministic unit vector u and unfold $H_{k+1}^{(w)} u$ as a sum over future states:

$$
H_{k+1}^{(w)} u = \sum_{l=1}^{n-k} g_{k,l}(Z_{k+l}), \quad g_{k,l}(z) = B^{n-k-l+1} \tilde{A}(z) B^{l-1} u. \tag{71}
$$

Each $g_{k,l}$ is centered under π (since $\pi\left(\tilde{A}\right) = 0$), and the two B-powers give the uniform bound

$$
\|g_{k,l}\|_\infty \le C(1-\alpha a)^{(n-k)/2}, \tag{72}
$$

which is independent of l. Squaring and summing over l produces only linear growth in $n-k$:

$$
\sum_{l=1}^{n-k} \|g_{k,l}\|_\infty^2 \le C(n-k)(1-\alpha a)^{n-k}. \tag{73}
$$

Note the contrast with the U_R analysis: there, UGE folded the l-sum into a single t_{mix} factor, whereas here the same l-independent bound is applied $(n-k)$ times — the price of conditional centering being weaker than π-centering.

Applying the Markov concentration lemma to the future chain conditionally on $\mathcal{F}_k$,

$$
\|H_{k+1}^{(w)} u - \mathbb{E}\left[H_{k+1}^{(w)} u \mid \mathcal{F}_k\right]\|_{L_p} \le C p^{1/2} t_{\mathrm{mix}}^{1/2} \sqrt{(n-k)(1-\alpha a)^{n-k}}, \tag{74}
$$

and multiplying by $\epsilon(Z_k)$ (which is $\mathcal{F}_k$-measurable and bounded by $\|\epsilon\|_\infty$) gives

$$
\|H_{k+1}^{(w)} \epsilon(Z_k) - v_k^{(w,\varepsilon)}(Z_k)\|_{L_p} \le C p^{1/2} t_{\mathrm{mix}}^{1/2} \ \|\epsilon\|_\infty \ \sqrt{(n-k)(1-\alpha a)^{n-k}}. \tag{75}
$$

The summands in U_M are centered conditionally on $\mathcal{F}_k$, but they are *not* forward martingale differences: the kernel $H_{k+1}^{(w)}$ peeks into the future of Z_k. To assemble the per-k bounds into a bound on the sum we therefore invoke the weighted Burkholder/block-coupling estimate for future-centered bilinear Markov sums of Samsonov et al. (2025, Proposition 9), specialized to the present constant-stepsize kernel:

$$
\|U_M\|_{L_p} \le C p \left(\sum_{k=1}^{n-1} \|H_{k+1}^{(w)} \epsilon(Z_k) - v_k^{(w,\varepsilon)}(Z_k)\|_{L_p}^2 \right)^{1/2}. \tag{76}
$$

Plugging in the previous bound and evaluating the resulting sum:

$$
\begin{aligned}
\|U_M\|_{L_p} &\le C p^{3/2} t_{\mathrm{mix}}^{1/2} \ \|\epsilon\|_\infty \ \left(\sum_{k=1}^{n-1} (n-k)(1-\alpha a)^{n-k} \right)^{1/2} && \text{(plug previous bound)} \\
&\le C p^{3/2} t_{\mathrm{mix}}^{1/2} \ \|\epsilon\|_\infty \ \frac{1}{\alpha a} && \left(\text{use} \sum_m m\, r^m \le C(1-r)^{-2}, r = 1-\alpha a \ \right).
\end{aligned} \tag{77}
$$

The extra factor $1/\sqrt{\alpha a}$ relative to U_R is precisely the cost of using conditional rather than stationary centering.

Step 3: assembly. Combining the two pieces via $S_n - \mathbb{E}S_n = -\alpha(U_M + U_R)$ and the triangle inequality,

$$\begin{aligned}\|S_n - \mathbb{E}S_n\|_{L_p} &\le \alpha \left(\|U_M\|_{L_p} + \|U_R\|_{L_p}\right) \\ &\le C\ \|\epsilon\|_\infty \left(p^{3/2} t_{\text{mix}}^{1/2} \frac{1}{a} + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\frac{\alpha}{a}}\right).\end{aligned} \tag{78}$$

The first term (from U_M) is the leading contribution for small α; the second (from U_R) carries the heavier t_{mix} dependence but vanishes as $\alpha \to 0$. Multiplying by α gives the asserted bound for $T_n^{(1,\alpha)} = -\alpha S_n$. □

3.2. Application to the PR-averaged RR Misadjustment

The PR-averaged Richardson–Romberg expansion produces, after Step (S8) of the Samsonov scheme applied separately at step sizes α and 2α, a "misadjustment" remainder

$$D_1^{\text{mis, RR}} = \frac{\sqrt{n}}{n - n_0} \sum_{k=n_0}^{n-1} \left(2J_k^{(1,\alpha)} - J_k^{(1,2\alpha)}\right), \tag{79}$$

whose centered part must be controlled to feed into a Berry–Esseen statement for $\sqrt{n}\left(\overline{\theta}_n^{(\text{RR},\alpha)} - \theta^*\right)$. We write

$$D_{1,\text{ c}}^{\text{mis, RR}} \coloneqq D_1^{\text{mis, RR}} - \mathbb{E}D_1^{\text{mis, RR}} \tag{80}$$

for this centered statistic.

The stationary bias is smaller than the fluctuation term. By Levin et al. (2025, Proposition 2),

$$\mathbb{E}_\pi\left[J_\infty^{(1,\alpha)}\right] = \alpha\Delta + O(\alpha^2), \tag{81}$$

so the linear term $\alpha\Delta$ cancels in the RR-combination and the per-iterate stationary RR bias is $O(\alpha^2)$. Therefore the PR-scaled stationary bias satisfies

$$\|\mathbb{E}D_1^{\text{mis, RR}}\| \le C\sqrt{n}\,\alpha^2. \tag{82}$$

What remains is the centered fluctuation.

Define

$$\Phi(p,\alpha) \coloneqq p^{3/2}\, \frac{t_{\text{mix}}^{1/2}}{a} + p^{1/2}\, t_{\text{mix}}^{3/2} \sqrt{\alpha/a}. \tag{83}$$

The lemma applied at α and at 2α gives, separately,

$$\|T_n^{(1,\alpha)} - \mathbb{E}T_n^{(1,\alpha)}\|_{L_p} \le C\alpha\,\Phi(p,\alpha), \quad \|T_n^{(1,2\alpha)} - \mathbb{E}T_n^{(1,2\alpha)}\|_{L_p} \le C\alpha\,\Phi(p,2\alpha) \le C'\alpha\,\Phi(p,\alpha), \tag{84}$$

where $C' = \sqrt{2}C$ absorbs the $\sqrt{2}$-factor coming from the 2α scaling. Combining the two by the triangle inequality and using the index-shift identity $T_k^{(1,w)} = \left(I - w\overline{A}\right) J_k^{(1,w)}$ together with the local inverse bound $\|\ \left(I - w\overline{A}\right)^{-1}\ \| \le 2$ for $w \in \{\alpha, 2\alpha\}$, we get

$$\|\left(2J_k^{(1,\alpha)} - J_k^{(1,2\alpha)}\right) - \mathbb{E}\left(2J_k^{(1,\alpha)} - J_k^{(1,2\alpha)}\right)\|_{L_p} \le C\alpha\,\Phi(p,\alpha), \tag{85}$$

uniformly in k. PR-averaging through $\frac{\sqrt{n}}{n-n_0}$ and absorbing the constant therefore yields

$$\|D_{1,\text{ c}}^{\text{mis, RR}}\|_{L_p} \le C\sqrt{n}\,\alpha\,\Phi(p,\alpha) = O\left(\sqrt{n}\,\alpha\right). \tag{86}$$

Together with the bias estimate,

$$\|D_1^{\text{mis, RR}}\|_{L_p} \le C\sqrt{n}\,\alpha\,\Phi(p,\alpha) + C\sqrt{n}\,\alpha^2. \tag{87}$$

At the optimal scale $\alpha \asymp n^{-1/2}$ the centered-fluctuation term is $O(1)$, whereas the stationary bias is $O\big(n^{-1/2}\big)$. Hence this depth-one route still does not yield a useful Berry–Esseen remainder of order $n^{-1/4}$: the centered misadjustment must be controlled more sharply to be subleading.

4. Richardson–Romberg PR Weight Bounds

Throughout this chapter, unnamed constants C may change from line to line and are independent of n, α, p, and q. Named constants display the dependencies that matter for the rate; fixed problem parameters may be absorbed only when they are not being tracked explicitly in the surrounding display.

4.1. PR-Averaged Error Decomposition and the RR Weight

The deterministic weight estimates start from the finite-start representation of $\sqrt{n}\Big(\overline{\theta}_n^{(\mathrm{RR},\alpha)} - \theta^*\Big)$ as a noise-weighted sum with a deterministic kernel. The stationary $n_0 = 0$ Berry–Esseen bound proved below is stated for the augmented-chain assembly built from the same full-window weights.

Depth-one expansion. Unfolding the error recursion of Chapter 1 gives, for every $k \geq 1$,

$$\theta_k^{(\alpha)} - \theta^* = -\alpha \sum_{l=1}^{k} \Gamma_{l+1:k}^{(\alpha)} \epsilon(Z_l) + \Gamma_{1:k}^{(\alpha)}(\theta_0 - \theta^*), \quad \Gamma_{l+1:k}^{(\alpha)} \coloneqq \prod_{j=l+1}^{k} \big(I - \alpha A(Z_j)\big). \tag{88}$$

Replacing each random product $\Gamma_{l+1:k}^{(\alpha)}$ by its deterministic counterpart $B_\alpha^{k-l} \coloneqq \Big(I - \alpha\overline{A}\Big)^{k-l}$ and absorbing the difference into a higher-order remainder yields the **depth-one decomposition** (Samsonov et al., 2025, Proposition 9):

$$\theta_k^{(\alpha)} - \theta^* = -\alpha \sum_{l=1}^{k} B_\alpha^{k-l} \epsilon(Z_l) + B_\alpha^{k}(\theta_0 - \theta^*) + R_k^{(\alpha)}, \tag{89}$$

where $R_k^{(\alpha)} \coloneqq J_k^{(1,\alpha)} + H_k^{(1,\alpha)}$ is the first misadjustment remainder. Its leading component $J_k^{(1,\alpha)}$ has a stationary bias of order α, so it is not treated as an $\alpha^{3/2}$ term. The Berry–Esseen proof below controls this remainder by refining it to depth two, where $J^{(2)} + H^{(2)}$ carries the $\alpha^{3/2}$ moment scale. The first sum is the **depth-zero** term and is the only piece that carries the limiting Gaussian.

PR averaging produces $\boldsymbol{Q_l^{(\alpha)}}$. Recall the PR average $\overline{\theta}_n^{(\alpha)} = (n - n_0)^{-1} \sum_{k=n_0}^{n-1} \theta_k^{(\alpha)}$. For notational clarity, and for the stationary result proved in this chapter, we set $n_0 = 0$ from this point on. A burned-in average has a different deterministic weight,

$$Q_{l,n_0}^{(\alpha)} = \frac{n}{n - n_0} \alpha \sum_{k=\max(n_0,l)}^{n-1} B_\alpha^{k-l}, \tag{90}$$

when the statistic is normalized as $-n^{-1/2} \sum_l Q_{l,n_0}^{(\alpha)} \epsilon(Z_l)$. The burned-in non-stationary theorem therefore requires separate weight, Poisson, and variance-comparison arguments. Subtracting θ^*, substituting the depth-one decomposition above, and **exchanging the order of summation** in the depth-zero piece,

$$\sum_{k=0}^{n-1} \sum_{l=1}^{k} B_\alpha^{k-l} \epsilon(Z_l) = \sum_{l=1}^{n-1} \left(\sum_{k=l}^{n-1} B_\alpha^{k-l} \right) \epsilon(Z_l), \tag{91}$$

isolates each noise sample $\epsilon(Z_l)$ together with the deterministic **PR weight**

$$Q_l^{(\alpha)} \coloneqq \alpha \sum_{k=l}^{n-1} B_\alpha^{k-l} = \alpha \sum_{j=0}^{n-l-1} B_\alpha^{j}. \tag{92}$$

Multiplying the resulting identity by $\sqrt{n}$ gives the PR-averaged decomposition

$$\sqrt{n}\left(\overline{\theta}_n^{(\alpha)} - \theta^*\right) = W^{(\alpha)} + D^{(\alpha)}, \tag{93}$$

where the **leading martingale-like sum** is

$$W^{(\alpha)} := -\frac{1}{\sqrt{n}} \sum_{l=1}^{n-1} Q_l^{(\alpha)}\, \epsilon(Z_l), \tag{94}$$

and the remainder $D^{(\alpha)}$ contains the deterministic transient $D_{\mathrm{tr}}^{(\alpha)}$ and the higher-order stochastic part $D_R^{(\alpha)}$:

$$D^{(\alpha)} := D_{\mathrm{tr}}^{(\alpha)} + D_R^{(\alpha)}, \quad D_{\mathrm{tr}}^{(\alpha)} := \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} B_\alpha^k(\theta_0 - \theta^*), \quad D_R^{(\alpha)} := \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} R_k^{(\alpha)}. \tag{95}$$

The first sum is the deterministic initial-condition transient. Under the full-average convention it is only $O\Big(\big(\sqrt{n}\alpha a\big)^{-1}\Big)$ in general, so it must either be retained explicitly or removed by a centered initialization $\theta_0 = \theta^*$. The second is the source of the leading non-Gaussian correction in the Berry–Esseen rate. Its RR-combination $2D_R^{(\alpha)} - D_R^{(2\alpha)}$ is the **misadjustment** $D_1^{\text{mis, RR}}$ controlled below by the Levin depth-two transfer.

RR combination produces $\mathcal{Q}_l^{\mathrm{RR}}$. The Richardson–Romberg iterate $\overline{\theta}_n^{(\mathrm{RR},\alpha)} := 2\overline{\theta}_n^{(\alpha)} - \overline{\theta}_n^{(2\alpha)}$ inherits the PR decomposition **by linearity**: applying the previous display at step sizes α and 2α separately and combining,

$$\sqrt{n}\left(\overline{\theta}_n^{(\mathrm{RR},\alpha)} - \theta^*\right) = W^{\mathrm{RR}} + D^{\mathrm{RR}}, \tag{96}$$

$$W^{\mathrm{RR}} := 2W^{(\alpha)} - W^{(2\alpha)} = -\frac{1}{\sqrt{n}} \sum_{l=1}^{n-1} \mathcal{Q}_l^{\mathrm{RR}}\, \epsilon(Z_l), \tag{97}$$

where the **RR weight** is

$$\mathcal{Q}_l^{\mathrm{RR}} := 2Q_l^{(\alpha)} - Q_l^{(2\alpha)}. \tag{98}$$

Since both PR averages share the **same** noise realization $\{Z_k\}$, the bracket $\mathcal{Q}_l^{\mathrm{RR}}$ is a single deterministic matrix kernel: all the stochastic content of W^{RR} now lives in $\epsilon(Z_l)$ alone.

Two deterministic estimates. The rest of the martingale approximation uses two quantities determined by $\mathcal{Q}_l^{\mathrm{RR}}$:

1. **Variance comparison.** The target CLT covariance of W^{RR} is $\Sigma_\infty = \overline{A}^{-1}\Sigma_\epsilon^{(\mathrm{M})}\overline{A}^{-\top}$ (the Markov-chain CLT covariance). Replacing each $\mathcal{Q}_l^{\mathrm{RR}}$ by the asymptotic weight $\overline{A}^{-1}$ gives this covariance, so the finite-n deviation is controlled by $\sum_l \|\mathcal{Q}_l^{\mathrm{RR}} - \overline{A}^{-1}\|^2$ — see Section 4.5.
2. **Poisson-equation / Abel-summation remainder.** The standard Berry–Esseen route for Markov-chain noise solves the Poisson equation $\hat{\epsilon} - \mathbf{Q}\hat{\epsilon} = \epsilon$, replaces $\epsilon(Z_l)$ by $\hat{\epsilon}(Z_l) - \hat{\epsilon}(Z_{l+1})$ (up to a martingale increment), and Abel-sums against the weight sequence $\mathcal{Q}_l^{\mathrm{RR}}$. The resulting remainder has norm bounded by the **total variation** $\sum_l \|\mathcal{Q}_{l+1}^{\mathrm{RR}} - \mathcal{Q}_l^{\mathrm{RR}}\|$.

Both quantities are estimated in Sections 4.3–4.4 below. The closed-form identities of Section 4.2 reduce them to elementary operator-norm calculations on $B_\alpha^m - B_{2\alpha}^m$.

4.2. Closed-Form Identities

Throughout this chapter write

$$B_\alpha := I - \alpha\overline{A}, \quad B_{2\alpha} := I - 2\alpha\overline{A}, \quad k := n - l. \tag{99}$$

We assume $\alpha, 2\alpha \in (0, \alpha_\infty]$, so that the Lyapunov contraction $\| B_\alpha^m \|_Q^2 \le (1 - \alpha a)^m$ and $\| B_{2\alpha}^m \|_Q^2 \le (1 - 2\alpha a)^m$ hold (and we use freely $1 - 2\alpha a \le 1 - \alpha a$).

The geometric-series identity $\sum_{j=0}^{m-1} B_\alpha^j = \left(\alpha\overline{A}\right)^{-1}(I - B_\alpha^m)$ converts Eq. 92 into the closed form

$$Q_l^{(\alpha)} = \alpha \sum_{j=0}^{n-l-1} B_\alpha^j = \alpha \left(\alpha \overline{A}\right)^{-1} \left(I - B_\alpha^{n-l}\right) = \overline{A}^{-1}\left(I - B_\alpha^{n-l}\right). \tag{100}$$

This makes the convergence $Q_l^{(\alpha)} \to \overline{A}^{-1}$ as $k = n - l \to \infty$ explicit: the only obstruction is the geometric Lyapunov tail B_α^k. Two immediate consequences are

$$Q_l^{(\alpha)} - \overline{A}^{-1} = -\overline{A}^{-1} B_\alpha^{n-l}, \quad Q_{l+1}^{(\alpha)} - Q_l^{(\alpha)} = -\alpha B_\alpha^{n-l-1}. \tag{101}$$

The first is the **asymptotic-weight error** and decays geometrically in k. The second is the **discrete derivative** used in Abel summation; note the explicit factor α in front.

Applying both identities at α and 2α and combining yields the basic **RR identities**:

$$\mathcal{Q}_l^{\mathrm{RR}} - \overline{A}^{-1} = -\overline{A}^{-1}\left(2B_\alpha^k - B_{2\alpha}^k\right), \tag{102}$$

$$\mathcal{Q}_{l+1}^{\mathrm{RR}} - \mathcal{Q}_l^{\mathrm{RR}} = -2\alpha\left(B_\alpha^{k-1} - B_{2\alpha}^{k-1}\right). \tag{103}$$

These are the exact expressions bounded in the next subsection. Note the structural asymmetry between them:

- The asymptotic-weight error $2B_\alpha^k - B_{2\alpha}^k$ is at the **single-trajectory** rate $(1 - \alpha a)^{k/2}$ — the triangle inequality $|2X - Y| \le 2\,|X| + |Y|$ does not see the RR coupling, and indeed the leading $-\overline{A}^{-1}$ in $\mathcal{Q}_l^{\mathrm{RR}}$ is the same as in the single-step weight $Q_l^{(\alpha)}$.
- The discrete derivative $B_\alpha^{k-1} - B_{2\alpha}^{k-1}$, however, is a **true** difference of contractions evaluated at the same exponent. The elementary identity $X^m - Y^m = (X - Y)\sum_{i=1}^m X^{i-1}Y^{m-i}$ with $X = B_\alpha$, $Y = B_{2\alpha}$, $X - Y = \alpha\overline{A}$ extracts an extra factor of α, gaining one full power of α over the single-step case $Q_{l+1}^{(\alpha)} - Q_l^{(\alpha)} = -\alpha B_\alpha^{k-1}$.

This local α-gain in the discrete derivative is the **only** manifestation of Richardson–Romberg cancellation visible at the level of the PR weights.

4.3. Pointwise Bounds for the RR Weights

Lemma 5. Let $\alpha, 2\alpha \in (0, \alpha_\infty]$, set $C_Q := \kappa_Q^{1/2} \parallel \overline{A}^{-1} \parallel$ and $\tilde{C}_A := \kappa_Q \parallel \overline{A} \parallel$, and write $k = n - l$.

(i) For every $1 \le l \le n-1$,

$$\parallel \mathcal{Q}_l^{\mathrm{RR}} - \overline{A}^{-1} \parallel \le 3C_Q(1 - \alpha a)^{k/2}. \tag{104}$$

(ii) For every $1 \le l \le n-2$,

$$\parallel \mathcal{Q}_{l+1}^{\mathrm{RR}} - \mathcal{Q}_l^{\mathrm{RR}} \parallel \le 2\tilde{C}_A\,\alpha^2\,(k-1)\,(1 - \alpha a)^{(k-2)/2}. \tag{105}$$

Proof of (i). Take norms in the basic RR identity above. Using the equivalence $\parallel \cdot \parallel \le \kappa_Q^{1/2} \parallel \cdot \parallel_Q$ and the Lyapunov contraction at both step sizes,

$$\parallel 2B_\alpha^k - B_{2\alpha}^k \parallel_Q \le 2 \parallel B_\alpha^k \parallel_Q + \parallel B_{2\alpha}^k \parallel_Q \le 2(1-\alpha a)^{k/2} + (1 - 2\alpha a)^{k/2} \le 3(1 - \alpha a)^{k/2}. \tag{106}$$

Submultiplicativity in $\parallel \cdot \parallel_Q$ followed by norm equivalence then gives

$$\parallel \mathcal{Q}_l^{\mathrm{RR}} - \overline{A}^{-1} \parallel \le \kappa_Q^{1/2} \parallel \overline{A}^{-1} \parallel \cdot \parallel 2B_\alpha^k - B_{2\alpha}^k \parallel_Q \le 3C_Q(1 - \alpha a)^{k/2}. \tag{107}$$

Proof of (ii). Apply the elementary identity $X^m - Y^m = (X - Y)\sum_{i=1}^m X^{i-1}Y^{m-i}$ with $X = B_\alpha$, $Y = B_{2\alpha}$, $X - Y = \alpha\overline{A}$, and $m = k - 1$:

$$B_\alpha^{k-1} - B_{2\alpha}^{k-1} = \alpha \overline{A} \sum_{i=1}^{k-1} B_\alpha^{i-1} \, B_{2\alpha}^{k-1-i}. \tag{108}$$

Each summand obeys, by submultiplicativity in $\| \cdot \|_Q$ and Lyapunov contraction,

$$\begin{aligned} \| \, B_\alpha^{i-1} \, B_{2\alpha}^{k-1-i} \, \| &\leq \kappa_Q^{1/2} \, (1-\alpha a)^{(i-1)/2} (1-2\alpha a)^{(k-1-i)/2} \\ &\leq \kappa_Q^{1/2} \, (1-\alpha a)^{(k-2)/2}, \end{aligned} \tag{109}$$

where in the last step we used $1 - 2\alpha a \leq 1 - \alpha a$. Summing over $i \in \{1, ..., k-1\}$ and absorbing constants,

$$\| \, B_\alpha^{k-1} - B_{2\alpha}^{k-1} \, \| \leq \alpha \kappa_Q \, \| \, \overline{A} \, \| \, (k-1)(1-\alpha a)^{(k-2)/2}. \tag{110}$$

Inserting this bound into the discrete-difference identity finishes the proof. □

4.4. Summed Bounds and Comparison with the Single-Step Case

Corollary 1. Under the assumptions of the previous lemma, uniformly in $n \geq 2$,

$$\sum_{l=1}^{n-1} \| \, Q_l^{\mathrm{RR}} - \overline{A}^{-1} \, \|^2 \leq \frac{C_1}{\alpha a}, \quad \sum_{l=1}^{n-2} \| \, Q_{l+1}^{\mathrm{RR}} - Q_l^{\mathrm{RR}} \, \| \leq \frac{C_2}{a^2}, \tag{111}$$

with $C_1 = 9 C_Q^2$ and $C_2 = 32 \tilde{C}_A$.

Proof. For the first sum apply (i) and the geometric series $\sum_{k \geq 1} (1-\alpha a)^k \leq \frac{1}{\alpha a}$. For the second, apply (ii) and

$$\sum_{k \geq 2} (k-1) \, (1-\alpha a)^{(k-2)/2} = \sum_{m \geq 0} (m+1)(1-\alpha a)^{m/2} \leq \frac{1}{\left(1 - \sqrt{1-\alpha a}\right)^2} \leq \frac{4}{(\alpha a)^2}, \tag{112}$$

where the last step uses $1 - \sqrt{1-\alpha a} \geq \alpha \frac{a}{2}$ for $\alpha a \leq 1/2$. Multiplying by $2 \tilde{C}_A \alpha^2$ gives the claim, up to the stated universal constant. □

4.5. Variance Comparison

After the Poisson decomposition the $l = 1$ noise sample is absorbed into the Poisson boundary remainder, while the martingale increments are indexed by $l \in \{2, ..., n-1\}$. The deterministic variance proxy used in the martingale Berry–Esseen step must therefore use the same index set:

$$\Sigma_n^{\mathrm{RR}} := \frac{1}{n} \sum_{l=2}^{n-1} Q_l^{\mathrm{RR}} \, \Sigma_\epsilon^{(\mathrm{M})} \left(Q_l^{\mathrm{RR}} \right)^\top, \quad \sigma_n^{2,\,\mathrm{RR}}(u) := u^\top \Sigma_n^{\mathrm{RR}} u, \tag{113}$$

with $\Sigma_\epsilon^{(\mathrm{M})}$ the symmetric long-run noise covariance of the Markov chain,

$$\Sigma_\epsilon^{(\mathrm{M})} = \mathbb{E}_\pi \left[\epsilon(Z_0) \epsilon(Z_0)^\top \right] + \sum_{j \geq 1} \left(\mathbb{E}_\pi \left[\epsilon(Z_0) \epsilon\left(Z_j\right)^\top \right] + \mathbb{E}_\pi \left[\epsilon\left(Z_j\right) \epsilon(Z_0)^\top \right] \right). \tag{114}$$

The asymptotic covariance is

$$\Sigma_\infty = \overline{A}^{-1} \Sigma_\epsilon^{(\mathrm{M})} \overline{A}^{-\top}, \quad \sigma^2(u) = u^\top \Sigma_\infty u. \tag{115}$$

The bounds of the previous section give a quantitative comparison between Σ_n^{RR} and Σ_∞.

Lemma 6. Let $\alpha, 2\alpha \in (0, \alpha_\infty]$, set $\Sigma \coloneqq \Sigma_\epsilon^{(\mathrm{M})}$, and assume $\|\ \Sigma\ \| < \infty$. Then

$$\|\ \Sigma_n^{\mathrm{RR}} - \Sigma_\infty\ \| \le \frac{C_3}{n\,\alpha a}, \tag{116}$$

with $C_3 = 12\,C_Q \|\ \overline{A}^{-1}\ \|\|\ \Sigma\ \| + 9\,C_Q^2 \|\ \Sigma\ \| + 2\|\ \Sigma_\infty\ \|$. Consequently, for every $u \in \mathbb{R}^d$,

$$|\ \sigma_n^{2,\,\mathrm{RR}}(u) - \sigma^2(u)\ | \le \frac{C_3 \|\ u\ \|^2}{n\,\alpha a}. \tag{117}$$

At the working scale $\alpha = c\,n^{-1/2}$ this is $O\big(n^{-1/2}\big)$.

Proof. Write $\Delta_l \coloneqq \mathcal{Q}_l^{\mathrm{RR}} - \overline{A}^{-1}$ and expand

$$\mathcal{Q}_l^{\mathrm{RR}}\,\Sigma\,\big(\mathcal{Q}_l^{\mathrm{RR}}\big)^\top - \Sigma_\infty = \underbrace{\Delta_l\,\Sigma\,\overline{A}^{-\top} + \overline{A}^{-1}\,\Sigma\,\Delta_l^\top}_{R_{1,l}} + \underbrace{\Delta_l\,\Sigma\,\Delta_l^\top}_{R_{2,l}}. \tag{118}$$

Submultiplicativity of the operator norm yields the pointwise bounds

$$\|\ R_{1,l}\ \| \le 2\|\ \overline{A}^{-1}\ \|\|\ \Sigma\ \|\|\ \Delta_l\ \|, \quad \|\ R_{2,l}\ \| \le \|\ \Sigma\ \|\|\ \Delta_l\ \|^2. \tag{119}$$

Summing the linear part with the bound from part (i) of the previous lemma and the geometric series $\sum_{k\ge 1}(1-\alpha a)^{k/2} \le \frac{1}{1-\sqrt{1-\alpha a}} \le \frac{2}{\alpha a}$ (valid for $\alpha a \le 1/2$),

$$\sum_{l=1}^{n-1}\|\ \Delta_l\ \| \le 3C_Q \sum_{k=1}^{n-1}(1-\alpha a)^{k/2} \le \frac{6C_Q}{\alpha a}. \tag{120}$$

The quadratic part is bounded directly by the previous corollary,

$$\sum_{l=1}^{n-1}\|\ \Delta_l\ \|^2 \le \frac{9C_Q^2}{\alpha a}. \tag{121}$$

Since the sum starts at $l = 2$, replacing every weight by $\overline{A}^{-1}$ gives $((n-2)/n)\Sigma_\infty$, not Σ_∞. Hence there is an additional deterministic finite-sum boundary term $2\Sigma_\infty/n$. Combining,

$$\begin{aligned}\|\ \Sigma_n^{\mathrm{RR}} - \Sigma_\infty\ \| &\le \frac{2\|\ \Sigma_\infty\ \|}{n} + \frac{1}{n}\sum_{l=2}^{n-1}\big(\|\ R_{1,l}\ \| + \|\ R_{2,l}\ \|\big) \\ &\le \frac{1}{n}\left(2\|\ \overline{A}^{-1}\ \|\|\ \Sigma\ \| \cdot \frac{6C_Q}{\alpha a} + \|\ \Sigma\ \| \cdot \frac{9C_Q^2}{\alpha a} + 2\|\ \Sigma_\infty\ \|/(\alpha a)\right) = \frac{C_3}{n\,\alpha a},\end{aligned} \tag{122}$$

where we used $\alpha a \le 1$. This proves the operator-norm bound. The scalar bound on $|\ \sigma_n^{2,\,\mathrm{RR}}(u) - \sigma^2(u)\ | = |\ u^\top(\Sigma_n^{\mathrm{RR}} - \Sigma_\infty)u\ |$ follows from the Cauchy–Schwarz inequality. □

4.6. Poisson Martingale Approximation

The variance comparison of the previous section identifies the limiting covariance of W^{RR}, but it does not by itself produce a martingale. This subsection converts W^{RR} into a martingale plus a quantitatively small remainder via the Poisson equation for the Markov chain $\{Z_l\}$. The deterministic kernel bounds of Sections 4.3–4.4 enter once and for all in the boundary/Abel control of the remainder.

Poisson kernel. Let Q denote the one-step Markov transition kernel of $(Z_k)_{k\ge 1}$, acting on bounded measurable functions $f : Z \to \mathbb{R}^d$ by

$$(Qf)(z) = \mathbb{E}[f(Z_{k+1})\ |\ Z_k = z]. \tag{123}$$

Under UGE 1 with mixing time t_{mix}, the geometric Dobrushin bound $\| Q^k \epsilon \|_\infty \le 2 \| \epsilon \|_\infty (1/4)^{\lfloor k/t_{\text{mix}} \rfloor}$ (valid for centered ϵ, $\pi(\epsilon) = 0$) makes the Poisson series

$$\hat{\epsilon} := \sum_{k=0}^{\infty} Q^k \epsilon \tag{124}$$

absolutely convergent in sup-norm, with

$$\| \hat{\epsilon} \|_\infty \le 2 \| \epsilon \|_\infty \sum_{k=0}^{\infty} (1/4)^{\lfloor k/t_{\text{mix}} \rfloor} \le 3\, t_{\text{mix}} \| \epsilon \|_\infty, \quad \| Q\hat{\epsilon} \|_\infty \le \| \hat{\epsilon} \|_\infty, \tag{125}$$

the second inequality because Q is a Markov kernel and contracts the sup-norm. By construction $\hat{\epsilon}$ solves the **Poisson equation**

$$\hat{\epsilon} - Q\hat{\epsilon} = \epsilon. \tag{126}$$

Conditional centering. Let $\mathcal{F}_l := \sigma(Z_1, \ldots, Z_l)$. Substituting Eq. 126 into the noise sample and adding and subtracting the conditional mean,

$$\epsilon(Z_l) = \underbrace{[\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1})]}_{\text{martingale increment}} + \underbrace{[Q\hat{\epsilon}(Z_{l-1}) - Q\hat{\epsilon}(Z_l)]}_{\text{telescope}}, \quad l \ge 2, \tag{127}$$

where the first bracket is centered conditionally on $\mathcal{F}_{l-1}$ since the Markov property gives $\mathbb{E}[\hat{\epsilon}(Z_l) \mid \mathcal{F}_{l-1}] = Q\hat{\epsilon}(Z_{l-1})$. The $l = 1$ term has no past state to condition on; we treat it directly via Eq. 126 as $\epsilon(Z_1) = \hat{\epsilon}(Z_1) - Q\hat{\epsilon}(Z_1)$. The decomposition therefore separates W^{RR} into a martingale piece and a deterministic-coefficient telescope; Abel summation against the kernel sequence $\{\mathcal{Q}_l^{\text{RR}}\}$ converts the telescope into boundary terms plus a sum against the discrete derivative $\mathcal{Q}_{l+1}^{\text{RR}} - \mathcal{Q}_l^{\text{RR}}$, which is exactly the object bounded in the Corollary of Section 4.4.

Lemma 7. Assume **UGE 1** and $\pi(\epsilon) = 0$. Set

$$\Delta M_l^{\text{RR}} := \mathcal{Q}_l^{\text{RR}} \left(\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1})\right), \quad 2 \le l \le n-1, \tag{128}$$

and let $M_n^{\text{RR}} := \sum_{l=2}^{n-1} \Delta M_l^{\text{RR}}$. Then $\{\Delta M_l^{\text{RR}}\}_{l=2}^{n-1}$ is a sequence of $\mathcal{F}_l$-martingale differences, and

$$W^{\text{RR}} = -\frac{1}{\sqrt{n}} M_n^{\text{RR}} + D_{2,n}^{\text{RR}}, \tag{129}$$

with the **Poisson boundary/Abel remainder**

$$D_{2,n}^{\text{RR}} := -\frac{1}{\sqrt{n}} \left[\mathcal{Q}_1^{\text{RR}}\, \hat{\epsilon}(Z_1) + \sum_{l=1}^{n-2} \left(\mathcal{Q}_{l+1}^{\text{RR}} - \mathcal{Q}_l^{\text{RR}}\right) Q\hat{\epsilon}(Z_l) \right]. \tag{130}$$

(The right boundary $\mathcal{Q}_{n-1}^{\text{RR}}\, Q\hat{\epsilon}(Z_{n-1})$ vanishes identically because $\mathcal{Q}_{n-1}^{\text{RR}} = 0$.) Moreover, with $C_{\mathcal{Q}} := \| \overline{A}^{-1} \| + 3C_Q$ a uniform bound on $\| \mathcal{Q}_l^{\text{RR}} \|$, and C_2 the constant from the Corollary of Section 4.4,

$$\| D_{2,n}^{\text{RR}} \|_\infty \le \frac{3\, t_{\text{mix}} \| \epsilon \|_\infty}{\sqrt{n}} \left(C_{\mathcal{Q}} + \frac{C_2}{a^2} \right). \tag{131}$$

Consequently, for every $p \ge 1$,

$$\| D_{2,n}^{\text{RR}} \|_{L_p} \le \frac{C\, t_{\text{mix}} \| \epsilon \|_\infty}{a^2 \sqrt{n}}, \tag{132}$$

with a constant C depending only on $\| \overline{A}^{-1} \|$, κ_Q, and $\| \overline{A} \|$.

Proof. The increments ΔM_l^{RR} are $\mathcal{F}_l$-martingale differences because $\mathcal{Q}_l^{\mathrm{RR}}$ is deterministic and the Markov property gives $\mathbb{E}[\hat{\epsilon}(Z_l) \mid \mathcal{F}_{l-1}] = Q\hat{\epsilon}(Z_{l-1})$.

For the decomposition, substitute Eq. 126 into the definition in Eq. 97:

$$W^{\mathrm{RR}} = -\frac{1}{\sqrt{n}} \sum_{l=1}^{n-1} \mathcal{Q}_l^{\mathrm{RR}} \left(\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_l)\right). \tag{133}$$

Peel off $l = 1$ and apply the conditional-centering identity above to the remaining $l \in \{2, ..., n-1\}$:

$$W^{\mathrm{RR}} = -\frac{1}{\sqrt{n}} \mathcal{Q}_1^{\mathrm{RR}} \, \hat{\epsilon}(Z_1) + \frac{1}{\sqrt{n}} \mathcal{Q}_1^{\mathrm{RR}} \, Q\hat{\epsilon}(Z_1) - \frac{1}{\sqrt{n}} M_n^{\mathrm{RR}} - \frac{1}{\sqrt{n}} \sum_{l=2}^{n-1} \mathcal{Q}_l^{\mathrm{RR}} \left(Q\hat{\epsilon}(Z_{l-1}) - Q\hat{\epsilon}(Z_l)\right). \tag{134}$$

Set $g_l := Q\hat{\epsilon}(Z_l)$ and Abel-sum the last sum:

$$\sum_{l=2}^{n-1} \mathcal{Q}_l^{\mathrm{RR}} (g_{l-1} - g_l) = \mathcal{Q}_2^{\mathrm{RR}} \, g_1 - \mathcal{Q}_{n-1}^{\mathrm{RR}} \, g_{n-1} + \sum_{l=2}^{n-2} \left(\mathcal{Q}_{l+1}^{\mathrm{RR}} - \mathcal{Q}_l^{\mathrm{RR}}\right) g_l. \tag{135}$$

Adding the contribution $\mathcal{Q}_1^{\mathrm{RR}} \, g_1$ from the peeled $l = 1$ term combines with $-\mathcal{Q}_2^{\mathrm{RR}} \, g_1$ to yield $-\left(\mathcal{Q}_2^{\mathrm{RR}} - \mathcal{Q}_1^{\mathrm{RR}}\right) g_1$, which folds the boundary $l = 1$ summand back into the discrete-derivative sum:

$$\begin{aligned} &-\mathcal{Q}_1^{\mathrm{RR}} \, \hat{\epsilon}(Z_1) + \mathcal{Q}_1^{\mathrm{RR}} \, g_1 - \mathcal{Q}_2^{\mathrm{RR}} \, g_1 + \mathcal{Q}_{n-1}^{\mathrm{RR}} \, g_{n-1} - \sum_{l=2}^{n-2} \left(\mathcal{Q}_{l+1}^{\mathrm{RR}} - \mathcal{Q}_l^{\mathrm{RR}}\right) g_l \\ &\quad = -\mathcal{Q}_1^{\mathrm{RR}} \, \hat{\epsilon}(Z_1) - \sum_{l=1}^{n-2} \left(\mathcal{Q}_{l+1}^{\mathrm{RR}} - \mathcal{Q}_l^{\mathrm{RR}}\right) g_l + \mathcal{Q}_{n-1}^{\mathrm{RR}} \, g_{n-1}. \end{aligned} \tag{136}$$

The rightmost boundary $\mathcal{Q}_{n-1}^{\mathrm{RR}} \, g_{n-1}$ vanishes because $\mathcal{Q}_{n-1}^{\mathrm{RR}} = 2\alpha I - 2\alpha I = 0$ identically (from $Q_{n-1}^{(\alpha)} = \alpha I$ and $Q_{n-1}^{(2\alpha)} = 2\alpha I$). Multiplying by $1/\sqrt{n}$ and absorbing the sign into $D_{2,n}^{\mathrm{RR}}$ produces exactly the stated formula.

For the sup-norm bound Eq. 131, use $\| \, g_l \, \|_\infty \le \| \, \hat{\epsilon} \, \|_\infty \le 3t_{\mathrm{mix}} \, \| \, \epsilon \, \|_\infty$, the uniform bound $\| \, \mathcal{Q}_1^{\mathrm{RR}} \, \| \le C_{\mathcal{Q}}$ on the left boundary, and the summed-total-variation bound $\sum_{l=1}^{n-2} \| \, \mathcal{Q}_{l+1}^{\mathrm{RR}} - \mathcal{Q}_l^{\mathrm{RR}} \, \| \le C_2/a^2$ on the Abel sum:

$$\| \, \sqrt{n} \, D_{2,n}^{\mathrm{RR}} \, \| \le 3t_{\mathrm{mix}} \, \| \, \epsilon \, \|_\infty \, \left(C_{\mathcal{Q}} + C_2/a^2\right). \tag{137}$$

The L_p bound is immediate from the deterministic sup-norm bound. □

4.7. Predictable Quadratic Variation Concentration

The Berry–Esseen step for the martingale M_n^{RR} requires a quantitative control of its predictable quadratic variation $\langle M^{\mathrm{RR}} \rangle_n$ around its deterministic asymptotic counterpart $n \, \sigma_n^{2,\,\mathrm{RR}}(u)$. This subsection produces such a control by applying the Markov concentration result of Section 2.2 to a suitable centered quadratic functional of the chain. The result is the RR analogue of Lemmas 22–23 of Samsonov et al. (2025).

Predictable quadratic variation. By the Poisson Martingale Approximation lemma above, the increments $\Delta M_l^{\mathrm{RR}} = \mathcal{Q}_l^{\mathrm{RR}} \left(\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1})\right)$ are $\mathcal{F}_l$-martingale differences. The Markov property $\mathbb{E}[\hat{\epsilon}(Z_l) \mid \mathcal{F}_{l-1}] = Q\hat{\epsilon}(Z_{l-1})$ gives, by direct computation,

$$\begin{aligned} \mathbb{E}\left[\Delta M_l^{\mathrm{RR}} \left(\Delta M_l^{\mathrm{RR}}\right)^\top \mid \mathcal{F}_{l-1}\right] &= \mathcal{Q}_l^{\mathrm{RR}} \mathbb{E}\left[\hat{\epsilon}(Z_l) \, \hat{\epsilon}(Z_l)^\top \mid \mathcal{F}_{l-1}\right] \left(\mathcal{Q}_l^{\mathrm{RR}}\right)^\top \\ &\quad - \mathcal{Q}_l^{\mathrm{RR}} \, Q\hat{\epsilon}(Z_{l-1}) \, (Q\hat{\epsilon})(Z_{l-1})^\top \left(\mathcal{Q}_l^{\mathrm{RR}}\right)^\top \\ &= \mathcal{Q}_l^{\mathrm{RR}} \, \mathcal{V}_\epsilon(Z_{l-1}) \left(\mathcal{Q}_l^{\mathrm{RR}}\right)^\top, \end{aligned} \tag{138}$$

where the cross-terms in the expansion of $(\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1}))(\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1}))^\top$ cancel by the Markov property and we have set

$$\mathcal{V}_\epsilon(z) := Q(\hat{\epsilon}\hat{\epsilon}^\top)(z) - (Q\hat{\epsilon})(z)\,(Q\hat{\epsilon})(z)^\top. \tag{139}$$

This is a matrix-valued conditional covariance of the Poisson martingale increment; the vector noise remains denoted by ϵ. Summing over $l \in \{2, ..., n-1\}$ and tracking that $\langle M^{\mathrm{RR}} \rangle_n := \sum_{l=2}^{n-1} \mathbb{E}\Big[\Delta M_l^{\mathrm{RR}}\left(\Delta M_l^{\mathrm{RR}}\right)^\top \mid \mathcal{F}_{l-1}\Big]$,

$$\langle M^{\mathrm{RR}} \rangle_n = \sum_{l=2}^{n-1} \mathcal{Q}_l^{\mathrm{RR}}\, \mathcal{V}_\epsilon(Z_{l-1})\left(\mathcal{Q}_l^{\mathrm{RR}}\right)^\top. \tag{140}$$

The conditional covariance function $\mathcal{V}_\epsilon$ has stationary mean equal to the long-run noise covariance,

$$\pi(\mathcal{V}_\epsilon) = \Sigma_\epsilon^{(\mathrm{M})} =: \Sigma, \tag{141}$$

which is a standard identity for the Poisson solution of a Markov chain (Samsonov et al. 2025, Eq. (10); see also Douc–Moulines–Priouret–Soulier 2018, Theorem 21.2.5).

Sup-norm bound on $\mathcal{V}_\epsilon$. Since $\|\, \hat{\epsilon} \,\|_\infty \le 3\, t_{\mathrm{mix}} \|\, \epsilon \,\|_\infty$ and Q is a Markov kernel (so Qf inherits the sup-norm of f),

$$\begin{aligned} \|\, \mathcal{V}_\epsilon \,\|_\infty &\le \|\, Q(\hat{\epsilon}\hat{\epsilon}^\top) \,\|_\infty + \|\, (Q\hat{\epsilon})(Q\hat{\epsilon})^\top \,\|_\infty \\ &\le 2 \|\, \hat{\epsilon} \,\|_\infty^2 \le 18\, t_{\mathrm{mix}}^2 \|\, \epsilon \,\|_\infty^2. \end{aligned} \tag{142}$$

Lemma 8. Assume **UGE 1** and $\pi(\epsilon) = 0$, with $\|\, \epsilon \,\|_\infty < \infty$. Let $C_{\mathcal{Q}}$ be the uniform bound on $\|\, \mathcal{Q}_l^{\mathrm{RR}} \,\|$ from the previous lemma. There exists a universal constant $C_4 > 0$ such that, for every $u \in \mathbb{R}^d$, every $p \ge 2$, every initial distribution ξ, and every $n \ge 2$,

$$\mathbb{E}_\xi^{1/p}[\,|\, u^\top \langle M^{\mathrm{RR}} \rangle_n u - n\, \sigma_n^{2,\,\mathrm{RR}}(u) \,|^p] \le C_4\, C_{\mathcal{Q}}^2 \|\, u \,\|^2 \|\, \epsilon \,\|_\infty^2\, t_{\mathrm{mix}}^{5/2} \sqrt{p\, n}. \tag{143}$$

Proof. Write $h_{l(z)} := u^\top \mathcal{Q}_l^{\mathrm{RR}}\, \mathcal{V}_\epsilon(z)\left(\mathcal{Q}_l^{\mathrm{RR}}\right)^\top u$ and $g_{l(z)} := h_{l(z)} - \pi(h_l)$. Submultiplicativity of the operator norm and Eq. 142 gives

$$|h_{l(z)}| \le C_{\mathcal{Q}}^2 \|\, u \,\|^2 \|\, \mathcal{V}_\epsilon \,\|_\infty \le 18\, C_{\mathcal{Q}}^2 \|\, u \,\|^2\, t_{\mathrm{mix}}^2 \|\, \epsilon \,\|_\infty^2, \quad \|\, g_l \,\|_\infty \le 2\, |h_{l(z)}| \le 36\, C_{\mathcal{Q}}^2 \|\, u \,\|^2\, t_{\mathrm{mix}}^2 \|\, \epsilon \,\|_\infty^2. \tag{144}$$

By Eq. 140 and the variance definition $n\, \sigma_n^{2,\,\mathrm{RR}}(u) = \sum_{l=2}^{n-1} \pi(h_l)$,

$$u^\top \langle M^{\mathrm{RR}} \rangle_n u - n\, \sigma_n^{2,\,\mathrm{RR}}(u) = \sum_{l=2}^{n-1} g_{l(Z_{l-1})}. \tag{145}$$

For the centered sum, set $\tilde{g}_i := g_{i+1}$ for $i \in \{1, ..., n-2\}$. By construction $\pi(\tilde{g}_i) = 0$ and $\|\, \tilde{g}_i \,\|_\infty \le c := 36\, C_{\mathcal{Q}}^2 \|\, u \,\|^2\, t_{\mathrm{mix}}^2 \|\, \epsilon \,\|_\infty^2$. The Markov concentration lemma of Section 2.2 (Levin et al. 2025, Lemma 11) applied to $\sum_{i=1}^{n-2} \tilde{g}_{i(Z_i)}$ yields, for every $t \ge 0$,

$$\mathbb{P}_\xi\left(\left|\sum_{i=1}^{n-2} \tilde{g}_{i(Z_i)}\right| \ge t\right) \le 2\exp\left(-\frac{t^2}{2\, u_n^2}\right), \quad u_n^2 \le 64\, t_{\mathrm{mix}}\,(n-2)\, c^2. \tag{146}$$

The sub-Gaussian-to-moment lemma of Section 2.2 then gives, for $p \ge 2$,

$$\mathbb{E}_\xi^{1/p}\left[\left|\sum_{l=2}^{n-1} g_{l(Z_{l-1})}\right|^p\right] \le 2^{1/p} \sqrt{p}\, u_n \le 2\sqrt{p}\, u_n \le 16\sqrt{2}\, \sqrt{p(n-2)}\, c\, \sqrt{t_{\mathrm{mix}}}. \tag{147}$$

Substituting c and bounding $\sqrt{n-2} \le \sqrt{n}$,

$$\mathbb{E}_\xi^{1/p}\left[\left|\sum_{l=2}^{n-1} g_{l(Z_{l-1})}\right|^p\right] \le 16\sqrt{2} \cdot 36\, C_{\mathcal{Q}}^2 \|\, u \,\|^2 \|\, \epsilon \,\|_\infty^2\, t_{\mathrm{mix}}^{5/2} \sqrt{p\, n} = 576\sqrt{2}\, C_{\mathcal{Q}}^2 \|\, u \,\|^2 \|\, \epsilon \,\|_\infty^2\, t_{\mathrm{mix}}^{5/2} \sqrt{p\, n}. \tag{148}$$

Rounding the resulting prefactor to $C_4 := 850$ gives the stated bound. □

Corollary 2. Under the assumptions of the previous lemma, for every $u \in \mathbb{R}^d$, every $p \geq 2$, and every $n \geq 2$,

$$\mathbb{E}_\xi^{1/p}[| \, u^\top \langle M^{\mathrm{RR}} \rangle_n u - n\, \sigma^2(u) \, |^p] \leq C_4\, C_{\mathcal{Q}}^2 \parallel u \parallel^2 \parallel \epsilon \parallel_\infty^2 t_{\mathrm{mix}}^{5/2} \sqrt{p\, n} + \frac{C_3 \parallel u \parallel^2}{\alpha\, a}, \tag{149}$$

with C_3 the variance-comparison constant of Section 4.5.

Proof. The triangle inequality and the variance-comparison lemma give

$$| \, u^\top \langle M^{\mathrm{RR}} \rangle_n u - n\, \sigma^2(u) \, | \leq | \, u^\top \langle M^{\mathrm{RR}} \rangle_n u - n\, \sigma_n^{2,\,\mathrm{RR}}(u) \, | + n \, | \, \sigma_n^{2,\,\mathrm{RR}}(u) - \sigma^2(u) \, |. \tag{150}$$

The first piece is bounded in L_p by Eq. 143; the second is the deterministic bound $n\, C_3 \parallel u \parallel^2 /(n\, \alpha\, a) = C_3 \parallel u \parallel^2 /(\alpha a)$. □

4.8. Martingale Berry–Esseen Step

The Poisson decomposition (Section 4.6) writes $W^{\mathrm{RR}} = -n^{-1/2} M_n^{\mathrm{RR}} + D_{2,n}^{\mathrm{RR}}$, with the remainder $D_{2,n}^{\mathrm{RR}}$ controlled in sup-norm at the order $n^{-1/2}$. The quadratic variation concentration (Section 4.7) bounds $|u^\top \langle M^{\mathrm{RR}} \rangle_n u - n\, \sigma_n^{2,\,\mathrm{RR}}(u)|$ in L_p by $C\sqrt{p\, n}$. This subsection assembles those two inputs into a Berry–Esseen rate for the **scalar** martingale $u^\top M_n^{\mathrm{RR}}$, the leading contribution to the final RR Berry–Esseen bound.

Bounded increments. Fix $u \in \mathbb{R}^d$ and set $X_l := u^\top \Delta M_l^{\mathrm{RR}}$ for $2 \leq l \leq n-1$. The X_l are $\mathcal{F}_l$-martingale differences (Section 4.6), and bounding the three factors $\parallel u \parallel$, $\parallel \mathcal{Q}_l^{\mathrm{RR}} \parallel \leq C_{\mathcal{Q}}$, and $\parallel \hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1}) \parallel \leq 2 \parallel \hat{\epsilon} \parallel_\infty \leq 6 t_{\mathrm{mix}} \parallel \epsilon \parallel_\infty$ yields the deterministic bound

$$|X_l| \leq \varkappa(u), \quad \varkappa(u) := 6\, t_{\mathrm{mix}}\, C_{\mathcal{Q}} \parallel \epsilon \parallel_\infty \parallel u \parallel. \tag{151}$$

The conditional variances $\sigma_l^2 := \mathbb{E}[X_l^2 \mid \mathcal{F}_{l-1}]$ sum to the predictable quadratic variation along u,

$$V_n^2 := \sum_{l=2}^{n-1} \sigma_l^2 = u^\top \langle M^{\mathrm{RR}} \rangle_n u. \tag{152}$$

Variance lower bound. Set $s_n^2 := n\, \sigma_n^{2,\,\mathrm{RR}}(u)$. The variance comparison lemma (Section 4.5) and assumption $\sigma^2(u) > 0$ give $|\sigma_n^{2,\,\mathrm{RR}}(u) - \sigma^2(u)| \leq C_3 \parallel u \parallel^2 /(n\alpha a)$, so whenever

$$n\, \alpha\, a \geq \frac{2\, C_3 \parallel u \parallel^2}{\sigma^2(u)} \tag{153}$$

one has $\sigma_n^{2,\,\mathrm{RR}}(u) \geq \sigma^2(u)/2$, i.e. $s_n^2 \geq n\sigma^2(u)/2$. At the working scale $\alpha = c\, n^{-1/2}$, the variance lower-bound condition Eq. 153 is satisfied for $n \geq \left(2C_3 \parallel u \parallel^2 /(c\, a\, \sigma^2(u))\right)^2$. The trivial upper bound $\sigma_n^{2,\,\mathrm{RR}}(u) \leq C_{\mathcal{Q}}^2 \parallel \Sigma_\epsilon^{(\mathrm{M})} \parallel \parallel u \parallel^2$ gives $s_n^2 \leq K^2(u)\, n$ with $K(u) := C_{\mathcal{Q}} \parallel \Sigma_\epsilon^{(\mathrm{M})} \parallel^{1/2} \parallel u \parallel$.

Bolthausen–Fan inequality. We apply Lemma 21 of Samsonov et al. (2025), which combines Bolthausen (1982) with Fan (2019). For any sequence of $\mathcal{F}_l$-martingale differences $\{X_l\}$ with $|X_l| \leq \varkappa$, any $p \geq 1$,

$$\begin{aligned} d_K(S_n/s_n, \mathcal{N}(0,1)) \leq\; & L_{B(\varkappa)} \frac{(2n+1)\log(2n+1)}{s_n^3} \\ & + C_1 \sqrt{p}\, s_n^{-2p/(2p+1)} \left(\mathbb{E} \mid V_n^2 - s_n^2 \mid^p\right)^{1/(2p+1)} \\ & + C_2\, s_n^{-2p/(2p+1)}\, p\, \varkappa^{2p/(2p+1)}, \end{aligned} \tag{154}$$

with $S_n := \sum_l X_l$, $L_{B(\varkappa)}$ the finite bounded-increment constant appearing in Samsonov et al.'s statement, and C_1, C_2 universal.

Theorem 1. Assume **UGE 1**, $\pi(\epsilon)=0$, $\|\,\epsilon\,\|_\infty<\infty$, $\sigma^2(u)>0$, and $\alpha, 2\alpha\in(0,\alpha_\infty]$. There exist constants $C_{K,1}(u), C_{K,2}(u)>0$ depending only on $\|\,u\,\|$, $\sigma(u)$, C_Q, t_{mix}, $\|\,\epsilon\,\|_\infty$, $\|\,\Sigma_\epsilon^{(\mathrm{M})}\,\|$, and the constants $L_{B(\varkappa(u))}, C_1, C_2$ of Eq. 154, such that for every $n\geq 3$ satisfying the variance lower-bound condition Eq. 153,

$$d_K\left(\frac{u^\top M_n^{\mathrm{RR}}}{\sqrt{n}\,\sigma_n^{\mathrm{RR}}(u)}, \mathcal{N}(0,1)\right)\leq \frac{C_{K,1}(u)\log^{3/4}n}{n^{1/4}}+\frac{C_{K,2}(u)\log n}{\sqrt{n}}. \tag{155}$$

Proof. Apply Eq. 154 with $S_n=u^\top M_n^{\mathrm{RR}}$, the increment bound $\varkappa=\varkappa(u)$ from Eq. 151, $s_n^2=n\,\sigma_n^{2,\,\mathrm{RR}}(u)$, and $p=\lceil\log n\rceil$ (so $\log n\leq p\leq\log n+1$). Bound the three terms in turn, using $s_n^2\in[n\,\sigma^2(u)/2, K^2(u)\,n]$ from the variance lower and upper bounds.

Term I (classical Bolthausen). From $s_n^3\geq(n\,\sigma^2(u)/2)^{3/2}$ and $(2n+1)\log(2n+1)\leq 3\,n\log n$ for $n\geq 3$,

$$L_{B(\varkappa(u))}\,\frac{(2n+1)\log(2n+1)}{s_n^3}\leq\frac{6\sqrt{2}\,L_{B(\varkappa(u))}}{\sigma^3(u)}\,\frac{\log n}{\sqrt{n}}=:\frac{C^{(\mathrm{I})}(u)\log n}{\sqrt{n}}. \tag{156}$$

Term III (Lindeberg). Write $a_p:=2p/(2p+1)=1-1/(2p+1)$. Using $s_n^{-a_p}=s_n^{-1}\,s_n^{1/(2p+1)}$ and $s_n\leq K(u)\sqrt{n}$,

$$s_n^{1/(2p+1)}\leq\max(1,K(u))^{1/(2p+1)}\,n^{1/(2(2p+1))}. \tag{157}$$

For $p\geq\log n$ one has $n^{1/(2(2p+1))}\leq n^{1/(4\log n)}=e^{1/4}$; similarly $\max(1,K(u))^{1/(2p+1)}\leq e^{1/2}$ once $n\geq\max(1,K(u))$, which is absorbed into $C^{(\mathrm{III})}(u)$. Bounding $\varkappa(u)^{a_p}\leq\max(1,\varkappa(u))$ and $s_n^{-1}\leq\sqrt{2}/(\sigma(u)\sqrt{n})$,

$$C_2\,s_n^{-a_p}\,p\,\varkappa(u)^{a_p}\leq\sqrt{2}\,e^{3/4}\,\frac{C_2\max(1,\varkappa(u))}{\sigma(u)}\,\frac{p}{\sqrt{n}}\leq\frac{C^{(\mathrm{III})}(u)\log n}{\sqrt{n}}, \tag{158}$$

with $C^{(\mathrm{III})}(u)$ depending on $\|\,u\,\|, \sigma(u), C_Q, \|\,\Sigma_\epsilon^{(\mathrm{M})}\,\|, t_{\mathrm{mix}}, \|\,\epsilon\,\|_\infty$ and on C_2.

Term II (conditional-variance concentration). By the bracket concentration bound above, for every $p\geq 2$,

$$\left(\mathbb{E}\mid V_n^2-s_n^2\mid^p\right)^{1/p}\leq B(u)\,\sqrt{p\,n},\quad B(u):=C_4\,C_Q^2\,\|\,u\,\|^2\,\|\,\epsilon\,\|_\infty^2\,t_{\mathrm{mix}}^{5/2}, \tag{159}$$

Using $s_n^{-a_p}=s_n^{-1}s_n^{1/(2p+1)}$, the variance lower bound $s_n^2\geq n\sigma^2\frac{u}{2}$, and the upper bound $s_n\leq K(u)\sqrt{n}$, the second Bolthausen–Fan term is bounded by

$$\begin{aligned}&C_1\,\sqrt{p}\,s_n^{-a_p}\left(\mathbb{E}\mid V_n^2-s_n^2\mid^p\right)^{1/(2p+1)}\\&\quad\leq C\,C_1\,\sigma^{-1}(u)\max(1,K(u))^{1/(2p+1)}\max(1,B(u))^{p/(2p+1)}\\&\qquad\times p^{(3p+1)/(2(2p+1))}n^{-p/(2(2p+1))}.\end{aligned} \tag{160}$$

For $p=\lceil\log n\rceil$ and n large enough depending on u, $p^{(3p+1)/(2(2p+1))}\leq C\log^{3/4}n$ and $n^{-p/(2(2p+1))}\leq Cn^{-1/4}$; the remaining $K(u)$- and $B(u)$-factors are absorbed into the direction-dependent constant. Thus

$$\text{Term II}\leq\frac{C^{(\mathrm{II})}(u)\log^{3/4}n}{n^{1/4}}. \tag{161}$$

Adding the three term bounds and setting $C_{K,1}(u):=C^{(\mathrm{II})}(u)$, $C_{K,2}(u):=C^{(\mathrm{I})}(u)+C^{(\mathrm{III})}(u)$ proves the martingale Berry–Esseen bound. □

Corollary 3. Under the hypotheses of the previous theorem,

$$d_K\left(\frac{u^\top M_n^{\mathrm{RR}}}{\sqrt{n}\,\sigma(u)}, \mathcal{N}(0,1)\right) \le \frac{C_{K,1}(u)\log^{3/4} n}{n^{1/4}} + \frac{C_{K,2}(u)\log n}{\sqrt{n}} + \frac{C_3 \parallel u \parallel^2}{n\,\alpha\,a\,\sigma^2(u)}. \tag{162}$$

At $\alpha = c\,n^{-1/2}$ the last term is $O\big(n^{-1/2}\big)$, hence absorbed into $C_{K,2}(u)\log n/\sqrt{n}$ up to a constant.

Proof. Set $r := \sigma_n^{\mathrm{RR}}(u)/\sigma(u)$ and $W := u^\top M_n^{\mathrm{RR}}/(\sqrt{n}\sigma_n^{\mathrm{RR}}(u))$. Then the statistic normalised by $\sigma(u)$ is Wr. Under the variance lower-bound condition Eq. 153, $r \ge 1/\sqrt{2}$, while the trivial upper bound on $\sigma_n^{2,\,\mathrm{RR}}(u)$ gives $r \le r_{\max(u)} < \infty$. On this compact interval the standard normal cdf satisfies

$$\sup_x |\Phi(x/r) - \Phi(x)| \le C_\Phi\,|r-1|, \quad C_\Phi := \sqrt{2}/\sqrt{\pi e}. \tag{163}$$

Therefore

$$d_{K(Wr,\mathcal{N}(0,1))} \le d_{K(W,\mathcal{N}(0,1))} + C_\Phi\,|r-1|. \tag{164}$$

The variance comparison of Section 4.5 gives

$$|r-1| = \frac{|\sigma_n^{2,\,\mathrm{RR}}(u) - \sigma^2(u)|}{\sigma(u)\,(\sigma_n^{\mathrm{RR}}(u) + \sigma(u))} \le \frac{C_3 \parallel u \parallel^2}{n\,\alpha\,a\,\sigma^2(u)}. \tag{165}$$

Adding this perturbation to the martingale Berry–Esseen bound proves the stated claim after absorbing C_Φ into the constants. □

This theorem concerns only the martingale term; the stationary assembly below adds the Poisson remainder and the Levin depth-two misadjustment.

4.9. Misadjustment via Levin Depth-Two

The Berry–Esseen assembly so far controls the leading martingale piece (Section 4.8) and the Poisson boundary remainder $D_{2,n}^{\mathrm{RR}}$ (Section 4.6). The remaining non-martingale contribution to $\sqrt{n}\,u^\top\Big(\overline{\theta}_n^{(\mathrm{RR},\alpha)} - \theta^*\Big)$ is the **PR-averaged misadjustment**

$$R_n^{\mathrm{mis,\,RR}} := \frac{1}{\sqrt{n}}\sum_{k=0}^{n-1}\Big(2R_k^{(\alpha)} - R_k^{(2\alpha)}\Big), \tag{166}$$

where $R_k^{(\alpha)} := J_k^{(1,\alpha)} + H_k^{(1,\alpha)}$ is the depth-one remainder of Eq. 89. A direct kernel-difference route only bounds $R_n^{\mathrm{mis,\,RR}}$ at order $O\big(\sqrt{n}\,\alpha\big) = O(1)$ at the working scale $\alpha \asymp n^{-1/2}$ — too crude for a Berry–Esseen rate $n^{-1/4}$. The depth-two route below transfers four statements of Levin et al. (2025) into the present notation and recovers the target $n^{-1/4}\,\mathrm{polylog}(n)$ rate.

Depth-two refinement. The deterministic-product expansion of Levin et al. extends the depth-one decomposition by one more level. For $\ell \ge 1$ define

$$J_n^{(\ell,\alpha)} := \Big(I - \alpha\overline{A}\Big)\,J_{n-1}^{(\ell,\alpha)} - \alpha\tilde{A}(Z_n)\,J_{n-1}^{(\ell-1,\alpha)}, \tag{167}$$

$$H_n^{(\ell,\alpha)} := (I - \alpha A(Z_n))\,H_{n-1}^{(\ell,\alpha)} - \alpha\tilde{A}(Z_n)\,J_{n-1}^{(\ell,\alpha)}, \tag{168}$$

with $J_0^{(\ell,\alpha)} = H_0^{(\ell,\alpha)} = 0$ and the $\ell = 0$ processes as in `last_iterate.typ`. Substituting $A(Z_n) = \overline{A} + \tilde{A}(Z_n)$ and grouping terms gives, by induction on n, the recursive identity

$$H_n^{(\ell,\alpha)} = J_n^{(\ell+1,\alpha)} + H_n^{(\ell+1,\alpha)}, \quad \ell \ge 0. \tag{169}$$

Applying Eq. 169 with $\ell = 1$ refines $R_k^{(\alpha)}$ to

$$R_k^{(\alpha)} = J_k^{(1,\alpha)} + J_k^{(2,\alpha)} + H_k^{(2,\alpha)}, \tag{170}$$

which splits the misadjustment into three structurally different pieces:

$$R_n^{\text{mis, RR}} = T_n^{(1)} + T_n^{(2)} + T_n^{(H)}, \tag{171}$$

$$T_n^{(j)} := \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} \Big(2J_k^{(j,\alpha)} - J_k^{(j,2\alpha)} \Big), \quad j \in \{1, 2\}, \tag{172}$$

with $T_n^{(H)}$ defined identically with $H^{(2)}$ in place of $J^{(j)}$. Each piece is bounded separately below.

Cited inputs. The depth-two route imports four statements of Levin et al. (2025), specialized to the constant step-size LSA setting. They are stated here for reference; their proofs are in the cited paper.

Lemma 9. (Levin Proposition 2 — stationary bias of $J^{(1)}$.) Under stationarity for the augmented chain $\Big(Z_{t+1}, J_t^{(0,\alpha)}, J_t^{(1,\alpha)}\Big)$,

$$\mathbb{E}_\pi \Big[J_\infty^{(1,\alpha)} \Big] = \alpha \, \Delta + R(\alpha), \quad \| \, R(\alpha) \, \| \le 12 \, \| \, \overline{A}^{-1} \, \| \, C_A^2 \, t_{\text{mix}}^2 \, \alpha^2 \, \| \, \epsilon \, \|_\infty, \tag{173}$$

with $\Delta := \overline{A}^{-1} \sum_{k \ge 1} \mathbb{E}_\pi \Big[\Big(Q^k \tilde{A} \Big)(Z_0) \, \epsilon(Z_0) \Big]$.

Lemma 10. (Levin Corollary 6 — centered bilinear L_p bound.) Define $\overline{\psi}_\alpha(j, z) := \tilde{A}(z) j - \mathbb{E}_{\Pi_{J^{(0)},\alpha}} \Big[\tilde{A}(Z_1) \, J_0^{(0,\alpha)} \Big]$. For every initial distribution, every $r \ge 1$, and every $p \ge 2$,

$$\left\| \sum_{t=0}^{r-1} \overline{\psi}_\alpha \Big(J_t^{(0,\alpha)}, Z_{t+1} \Big) \right\|_{L_p} \le c_{W,1} \, p^{3/2} \sqrt{\alpha r} + c_{W,2} \, p^3 \, \alpha^{-1/2} \log^{1/p}(1/(\alpha a)), \tag{174}$$

with $c_{W,1}, c_{W,2}$ depending only on $C_A, \kappa_Q, t_{\text{mix}}, \| \, \epsilon \, \|_\infty$. The precise logarithmic factor is not rate-critical below; it is absorbed into $\operatorname{polylog}(n)$ in the working-scale corollaries.

Lemma 11. (Levin Propositions 8 and 9 — high-order moment bounds.) For every $q \ge 2$ and every p satisfying $2 \le p \le q/2$, under the step-size restriction $\alpha \le \alpha_*(q, t_{\text{mix}})$ of Levin et al. (2025),

$$\| \, J_n^{(2,\alpha)} \, \|_{L_p} \le D_J \, t_{\text{mix}}^{5/2} \, p^{7/2} \, \alpha^{3/2} \log^{3/2}(1/(\alpha a)), \tag{175}$$

$$\| \, H_n^{(2,\alpha)} \, \|_{L_p} \le D_H \, d^{1/q} \, t_{\text{mix}}^{5/2} \, p^{7/2} \, \alpha^{3/2} \log^{3/2}(1/(\alpha a)), \tag{176}$$

uniformly in n, with D_J, D_H depending only on $C_A, \kappa_Q, \| \, \overline{A}^{-1} \, \|, \| \, \epsilon \, \|_\infty$.

Stationary augmented-chain convention. The recursions above are displayed in finite-time notation with $J_0^{(\ell,\alpha)} = H_0^{(\ell,\alpha)} = 0$, while Levin Proposition 2 is a statement about the stationary augmented chain. The misadjustment estimates below use the stationary versions of $\Big(Z_{t+1}, J_t^{(0,w)}, J_t^{(1,w)}, J_t^{(2,w)}, H_t^{(2,w)}\Big)$, $w \in \{\alpha, 2\alpha\}$, as deterministic centers and should be read under this stationary augmented-chain convention. We keep the same symbols $J_t^{(\ell,w)}$ and $H_t^{(\ell,w)}$ to avoid duplicating notation. For a zero-start full average, the transfer from finite-start sums to stationary centered sums is not a single terminal ρ^n term: summing pointwise contractions over $k = 0, \dots, n-1$ produces a startup contribution of order

$$\frac{1}{\sqrt{n}} \sum_{k \ge 0} \rho_\alpha^k \asymp \frac{1}{\sqrt{n} \, \alpha a} \tag{177}$$

when $\rho_\alpha \approx \exp(-c\alpha a)$. This term is not part of the theorem proved here. A practical arbitrary-start theorem requires either a genuine burn-in average with the burned-in weights $Q_{l,n_0}^{(\alpha)}$ of Section 4.1 or a separate transfer lemma with the correct accumulated startup dependence.

Lemma 12. (Stationary-limit transfer for the centered first-order iterate.) Fix $w \in (0, \alpha_\infty]$ and set $B_w := I - w\overline{A}$. Assume the stationary augmented-chain convention above, **UGE 1**, $\pi\left(\tilde{A}\right) = 0$, and $\| \epsilon \|_\infty < \infty$. Let $\left(J_t^{(0,w)}, J_t^{(1,w)}\right)_{t\in\mathbb{Z}}$ be the stationary two-sided solution of the first two perturbation recursions, and set $T_t^{(1,w)} := B_w\, J_t^{(1,w)}$. Then, with

$$\Phi(p, w) := p^{3/2}\, t_{\mathrm{mix}}^{1/2}/a + p^{1/2}\, t_{\mathrm{mix}}^{3/2}\sqrt{w/a}, \tag{178}$$

there exists a constant $C_{\mathrm{stat},1}$, depending only on the constants in the zero-start centered last-iterate bound of `last_iterate.typ`, such that for every $p \geq 2$ and every $t \in \mathbb{Z}$,

$$\| T_t^{(1,w)} - \mathbb{E}_\pi T_t^{(1,w)} \|_{L_p} \leq C_{\mathrm{stat},1}\, w\, \Phi(p, w). \tag{179}$$

Consequently, whenever $\|B_w^{-1}\| \leq 2$,

$$\| J_t^{(1,w)} - \mathbb{E}_\pi J_t^{(1,w)} \|_{L_p} \leq 2C_{\mathrm{stat},1}\, w\, \Phi(p, w). \tag{180}$$

Proof. For $m \geq 1$, run the same recursions on the stationary driving chain over the finite window $\{t - m + 1, ..., t\}$ with zero initial conditions at time $t - m$; denote the resulting variables by $J_{t,m}^{(0,w)}$ and $J_{t,m}^{(1,w)}$. Stationarity of the driving chain implies that $B_w J_{t,m}^{(1,w)}$ has the same law as the zero-start $T_m^{(1,w)}$ in `last_iterate.typ`. Therefore the centered last-iterate bound gives, uniformly in m and t,

$$\| B_w J_{t,m}^{(1,w)} - \mathbb{E} B_w J_{t,m}^{(1,w)} \|_{L_p} \leq C_{\mathrm{stat},1}\, w\, \Phi(p, w). \tag{181}$$

The deterministic-product expansions give

$$J_{t,m}^{(0,w)} = -w \sum_{r=0}^{m-1} B_w^r \epsilon(Z_{t-r}), \tag{182}$$

and express $J_{t,m}^{(1,w)}$ as a bounded double series whose summands are dominated by $Cw^2 s(1 - wa)^{s/2}\, \|\epsilon\|_\infty$ at total lag s. Since $\sum_{s\geq 0} s(1 - wa)^{s/2} < \infty$, the finite-past approximations are Cauchy in every L_p and converge to the stationary two-sided solution. Passing to the limit in the preceding uniform bound yields the displayed estimate for $T_t^{(1,w)}$. The estimate for $J_t^{(1,w)}$ then follows from $J_t^{(1,w)} - \mathbb{E}_\pi J_t^{(1,w)} = B_w^{-1}\Big(T_t^{(1,w)} - \mathbb{E}_\pi T_t^{(1,w)}\Big)$. $\square$

Telescoping identity for $J^{(1)}$. Summing the recursion $J_k^{(1,\alpha)} = \left(I - \alpha\overline{A}\right) J_{k-1}^{(1,\alpha)} - \alpha\tilde{A}(Z_k)\, J_{k-1}^{(0,\alpha)}$ from $k = 1$ to n and rearranging gives, for an arbitrary initial value,

$$\overline{A} \sum_{k=0}^{n-1} J_k^{(1,\alpha)} = -\sum_{k=1}^{n} \tilde{A}(Z_k)\, J_{k-1}^{(0,\alpha)} + \frac{1}{\alpha}\left(J_0^{(1,\alpha)} - J_n^{(1,\alpha)}\right). \tag{183}$$

The stationary version of the same recursion yields $\mathbb{E}_\pi\Big[\tilde{A}(Z_1)\, J_0^{(0,\alpha)}\Big] = -\overline{A}\, \mathbb{E}_\pi\Big[J_\infty^{(1,\alpha)}\Big]$. Subtracting $n\, \mathbb{E}_\pi\Big[J_\infty^{(1,\alpha)}\Big]$ from both sides and applying $\overline{A}^{-1}$ gives the **centered telescoping identity**

$$\sum_{k=0}^{n-1}\left(J_k^{(1,\alpha)} - \mathbb{E}_\pi\Big[J_\infty^{(1,\alpha)}\Big]\right) = -\overline{A}^{-1} \sum_{k=1}^{n} \overline{\psi}_\alpha\left(J_{k-1}^{(0,\alpha)}, Z_k\right) + \frac{1}{\alpha}\, \overline{A}^{-1}\left(J_0^{(1,\alpha)} - J_n^{(1,\alpha)}\right). \tag{184}$$

This identity is the bridge between the (vector-valued) PR-average of $J^{(1)}$ and the centered bilinear sum bounded in Levin Corollary 6.

Lemma 13. Assume the stationary augmented-chain convention above, **UGE 1**, $\pi\left(\tilde{A}\right)=0$, $\pi(\epsilon)=0$, $\| \epsilon \|_\infty < \infty$, and $\alpha, 2\alpha \in (0, \alpha_\infty]$. Set

$$\Phi(p,\alpha) := p^{3/2}\, t_{\mathrm{mix}}^{1/2}/a + p^{1/2}\, t_{\mathrm{mix}}^{3/2}\sqrt{\alpha/a}. \tag{185}$$

There exists a constant $C_{\mathrm{mis},1}$ depending on $\| \overline{A} \|, \| \overline{A}^{-1} \|, \kappa_Q, C_A, \| \epsilon \|_\infty, t_{\mathrm{mix}}$, the universals $c_{W,1}, c_{W,2}$ of Levin Corollary 6, and the constant of the centered bound in `last_iterate.typ`, such that for every $p \geq 2$ and every $n \geq 2$,

$$\begin{aligned}\| T_n^{(1)} \|_{L_p} &\leq C_{\mathrm{mis},1}\sqrt{n}\,\alpha^2 + C_{\mathrm{mis},1}\, p^{3/2}\sqrt{\alpha} \\ &\quad + C_{\mathrm{mis},1}\, p^3\,(\alpha n)^{-1/2}\log^{1/p}(1/(\alpha a)) + C_{\mathrm{mis},1}\,\Phi(p,\alpha)\, n^{-1/2}.\end{aligned} \tag{186}$$

Proof. Decompose $T_n^{(1)} = T_n^{(1,\,\mathrm{b})} + T_n^{(1,\,\mathrm{c})}$ via the bias-fluctuation split

$$T_n^{(1,\,\mathrm{b})} := \sqrt{n}\left(2\,\mathbb{E}_\pi\left[J_\infty^{(1,\alpha)}\right] - \mathbb{E}_\pi\left[J_\infty^{(1,2\alpha)}\right]\right), \tag{187}$$

$$T_n^{(1,\,\mathrm{c})} := \frac{1}{\sqrt{n}}\sum_{k=0}^{n-1}\sum_{w\in\{\alpha,2\alpha\}} c_w\left(J_k^{(1,w)} - \mathbb{E}_\pi\left[J_\infty^{(1,w)}\right]\right), \quad c_\alpha = 2, c_{2\alpha} = -1. \tag{188}$$

The bias is deterministic; the centered piece carries all the L_p fluctuation.

Bias. By Levin Proposition 2 applied at $w \in \{\alpha, 2\alpha\}$, $\mathbb{E}_\pi\left[J_\infty^{(1,w)}\right] = w\,\Delta + R(w)$ with $\| R(w) \| \leq 12 \| \overline{A}^{-1} \| C_A^2\, t_{\mathrm{mix}}^2\, w^2 \| \epsilon \|_\infty$. The leading $w\,\Delta$ cancels in the RR combination, $2\alpha\,\Delta - 2\alpha\,\Delta = 0$, leaving

$$\| 2\,\mathbb{E}_\pi\left[J_\infty^{(1,\alpha)}\right] - \mathbb{E}_\pi\left[J_\infty^{(1,2\alpha)}\right] \| \leq 2 \| R(\alpha) \| + \| R(2\alpha) \| \leq 72 \| \overline{A}^{-1} \| C_A^2\, t_{\mathrm{mix}}^2\, \alpha^2 \| \epsilon \|_\infty. \tag{189}$$

Multiplying by $\sqrt{n}$ produces the first term of the bound.

Centered. Apply Eq. 184 at $w \in \{\alpha, 2\alpha\}$, take L_p norms, and combine via the triangle inequality:

$$\begin{aligned}\| T_n^{(1,\,\mathrm{c})} \|_{L_p} \leq \frac{1}{\sqrt{n}}\sum_{w\in\{\alpha,2\alpha\}} |c_w| \Bigg(&\| \overline{A}^{-1} \| \cdot \| \sum_{k=1}^{n} \overline{\psi}_w\left(J_{k-1}^{(0,w)}, Z_k\right) \|_{L_p} \\ &+ \frac{1}{w} \| \overline{A}^{-1} \| \cdot \| J_0^{(1,w)} - J_n^{(1,w)} \|_{L_p}\Bigg).\end{aligned} \tag{190}$$

For the bilinear sum apply Levin Corollary 6 with $r = n$,

$$\| \sum_{k=1}^{n} \overline{\psi}_w \|_{L_p} \leq c_{W,1}\, p^{3/2}\sqrt{wn} + c_{W,2}\, p^3\, w^{-1/2}\log^{1/p}(1/(wa)), \tag{191}$$

and divide by $\sqrt{n}$, using $w \in [\alpha, 2\alpha]$ to upper-bound by the α-form,

$$\frac{1}{\sqrt{n}} \| \sum \overline{\psi}_w \|_{L_p} \leq \sqrt{2}c_{W,1}\, p^{3/2}\sqrt{\alpha} + \sqrt{2}c_{W,2}\, p^3\,(\alpha n)^{-1/2}\log^{1/p}(1/(\alpha a)). \tag{192}$$

For the boundary term, apply the stationary-limit transfer Lemma 12 to $T_n^{(1,w)} = \left(I - w\overline{A}\right) J_n^{(1,w)}$ with $\| \left(I - w\overline{A}\right)^{-1} \| \leq 2$ for $w \leq \alpha_\infty$. This gives

$$\| J_n^{(1,w)} - \mathbb{E}J_n^{(1,w)} \|_{L_p} \leq Cw\,\Phi(p,w), \tag{193}$$

and $\| \mathbb{E}J_n^{(1,w)} \| \leq w \| \Delta \| + \| R(w) \| \leq Cw$ by Levin Proposition 2; combining,

$$\| J_n^{(1,w)} \|_{L_p} \leq Cw(\Phi(p,w) + 1) \leq Cw\,\Phi(p,w) \quad \text{for } p \geq 2. \tag{194}$$

The same bound holds for $J_0^{(1,w)}$ under stationarity, so $w^{-1} \parallel J_0^{(1,w)} - J_n^{(1,w)} \parallel_{L_p} \leq C\,\Phi(p,w) \leq \sqrt{2}\,C\,\Phi(p,\alpha)$ after absorbing the factor 2 into C. Dividing by $\sqrt{n}$ produces the last term. Summing the three RR-combined pieces and absorbing universal factors into a single $C_{\text{mis},1}$ completes the proof. □

Lemma 14. Under the assumptions of the previous lemma, for every $p \geq 2$ and every $q \geq 2$ satisfying $p \leq q/2$ and $2\alpha \leq \alpha_*(q, t_{\text{mix}})$,

$$\parallel T_n^{(2)} \parallel_{L_p} + \parallel T_n^{(H)} \parallel_{L_p} \leq C_{\text{mis},2} \left(1 + d^{1/q}\right) p^{7/2}\, t_{\text{mix}}^{5/2} \sqrt{n}\, \alpha^{3/2} \log^{3/2}(1/(\alpha a)), \tag{195}$$

with $C_{\text{mis},2} \coloneqq 6\sqrt{2}^3\,(D_J + D_H)$.

Proof. Triangle inequality on the n summands of $T_n^{(2)}$, then Levin Propositions 8 and 9 applied uniformly in k and $w \in \{\alpha, 2\alpha\}$:

$$\begin{aligned} \parallel T_n^{(2)} \parallel_{L_p} &\leq \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} \sum_{w \in \{\alpha, 2\alpha\}} |c_w| \parallel J_k^{(2,w)} \parallel_{L_p} \\ &\leq 3\sqrt{n} \sup_{w \in \{\alpha,2\alpha\}} \parallel J_k^{(2,w)} \parallel_{L_p} \\ &\leq 3\, D_J\, (2\alpha)^{3/2}\, t_{\text{mix}}^{5/2}\, p^{7/2} \sqrt{n} \log^{3/2}(1/(\alpha a)), \end{aligned} \tag{196}$$

where $|c_\alpha| + |c_{2\alpha}| = 3$ absorbs the RR combination and $(2\alpha)^{3/2}$ the worse bound at the larger step. Identical argument for $T_n^{(H)}$, with the additional $d^{1/q}$ factor of Levin Proposition 9. Adding the two bounds gives the lemma. □

Theorem 2. (Stationary PR-averaged RR misadjustment bound.) Assume **UGE 1**, $\pi\left(\tilde{A}\right) = 0$, $\pi(\epsilon) = 0$, $\parallel \epsilon \parallel_\infty < \infty$, $\alpha, 2\alpha \in (0, \alpha_\infty]$, and the stationary augmented-chain convention above. There exists a constant C depending on the universal and problem constants of the previous two lemmas such that for every $p \geq 2$, every $q \geq 2$ satisfying $p \leq q/2$, every $2\alpha \leq \alpha_*(q, t_{\text{mix}})$, and every $n \geq 2$,

$$\begin{aligned} \parallel R_n^{\text{mis, RR}} \parallel_{L_p} &\leq C\sqrt{n}\,\alpha^2 + C\left(1 + d^{1/q}\right) p^{7/2}\, t_{\text{mix}}^{5/2} \sqrt{n}\, \alpha^{3/2} \log^{3/2}(1/(\alpha a)) \\ &\quad + C\, p^{3/2} \sqrt{\alpha} + C\, p^3\, (\alpha n)^{-1/2} \log^{1/p}(1/(\alpha a)) + C\,\Phi(p,\alpha)\, n^{-1/2}. \end{aligned} \tag{197}$$

Proof. Triangle inequality on Eq. 171: $\parallel R_n^{\text{mis, RR}} \parallel_{L_p} \leq \parallel T_n^{(1)} \parallel_{L_p} + \parallel T_n^{(2)} \parallel_{L_p} + \parallel T_n^{(H)} \parallel_{L_p}$. Combine the centered $T^{(1)}$ bound and the raw $T^{(2)} + T^{(H)}$ bound. □

Corollary 4. At the working scale $\alpha = c\, n^{-1/2}$, with $p = \max(2, \lceil \log n \rceil)$ and $q = \max(2p, \lceil \log(e\, d) \rceil, 2)$, and for n large enough that $2\alpha \leq \alpha_*(q, t_{\text{mix}})$,

$$\parallel R_n^{\text{mis, RR}} \parallel_{L_p} \leq C \operatorname{polylog}(n)\, n^{-1/4}, \tag{198}$$

matching the leading martingale Berry–Esseen rate of Section 4.8.

Proof. At $\alpha = c\, n^{-1/2}$: $\sqrt{n}\alpha^2 = c^2\, n^{-1/2}$, $\sqrt{n}\alpha^{3/2} = c^{3/2}\, n^{-1/4}$, $\sqrt{\alpha} = c^{1/2}\, n^{-1/4}$, $(\alpha n)^{-1/2} = c^{-1/2}\, n^{-1/4}$, and $\Phi(p,\alpha)\, n^{-1/2} = O\left(p^{3/2}\, n^{-1/2}\right)$. With $p \asymp \log n$ and $q \geq \log(ed)$, the factor $d^{1/q}$ is bounded by a universal constant, and the dominant order in the stationary misadjustment bound is $\operatorname{polylog}(n)\, n^{-1/4}$. □

The theorem just proved is only a stationary augmented-chain misadjustment bound at $n_0 = 0$. A finite-start theorem with burn-in requires the burned-in weights $Q_{l,n_0}^{(\alpha)}$ and is not used in the assembly below.

4.10. Smoothing Assembly

The previous sections produced the four ingredients of the stationary $n_0 = 0$ Berry–Esseen program for the PR-averaged Richardson–Romberg statistic:

1. the **Poisson decomposition** $W^{\mathrm{RR}} = -n^{-1/2} M_n^{\mathrm{RR}} + D_{2,n}^{\mathrm{RR}}$ with deterministic sup-norm control of $D_{2,n}^{\mathrm{RR}}$ at the order $t_{\mathrm{mix}}/(a^2\sqrt{n})$;
2. the **predictable quadratic variation concentration** of $u^\top \langle M^{\mathrm{RR}} \rangle_n u$ around $n\,\sigma_n^{2,\,\mathrm{RR}}(u)$ in L_p at the order $\sqrt{p\,n}$;
3. the **martingale Berry–Esseen** for $u^\top M_n^{\mathrm{RR}}/(\sqrt{n}\,\sigma_n^{\mathrm{RR}}(u))$ at rate $\log^{3/4}(n)\,n^{-1/4}$;
4. the **Levin depth-two misadjustment bound** on $R_n^{\mathrm{mis,\,RR}}$ controlling the non-martingale residual at the same $\log^{c(n)} n^{-1/4}$ rate.

The smoothing inequality of Bobkov–Götze (Samsonov et al. 2025, Proposition 12) assembles these four into a single Kolmogorov bound on the stationary $n_0 = 0$ augmented-chain RR statistic.

Stationary assembled statistic. Combining the depth-one identity in Section 4.1 with the Poisson decomposition of Section 4.6 gives a finite-start identity. Since the transfer from a zero-start recursion to the stationary augmented chain is not proved here, the bound below is stated for the stationary $n_0 = 0$ assembled scalar statistic

$$S_{n,\,\mathrm{stat}}^{\mathrm{RR}}(u) := -\frac{u^\top M_n^{\mathrm{RR}}}{\sqrt{n}} + u^\top \mathcal{R}_{n,\,\mathrm{stat}}^{\mathrm{RR}}, \tag{199}$$

$$\mathcal{R}_{n,\,\mathrm{stat}}^{\mathrm{RR}} := D_{2,n}^{\mathrm{RR}} + R_n^{\mathrm{mis,\,RR}}, \tag{200}$$

where $D_{2,n}^{\mathrm{RR}}$ is the Poisson boundary/Abel remainder of Section 4.6 and $R_n^{\mathrm{mis,\,RR}}$ is the Levin depth-two misadjustment defined in Eq. 166, both read under the stationary augmented-chain convention. The deterministic-start transient $D_{\mathrm{tr}}^{\mathrm{RR}} := 2\,D_{\mathrm{tr}}^{(\alpha)} - D_{\mathrm{tr}}^{(2\alpha)}$ is not part of this stationary result; handling it together with the accumulated startup error belongs to a later finite-start/burn-in transfer theorem. Dividing by $\sigma_n^{\mathrm{RR}}(u)$,

$$\frac{S_{n,\,\mathrm{stat}}^{\mathrm{RR}}(u)}{\sigma_n^{\mathrm{RR}}(u)} = X_n + Y_n, \tag{201}$$

$$X_n := -\frac{u^\top M_n^{\mathrm{RR}}}{\sqrt{n}\,\sigma_n^{\mathrm{RR}}(u)}, \quad Y_n := \frac{u^\top \mathcal{R}_{n,\,\mathrm{stat}}^{\mathrm{RR}}}{\sigma_n^{\mathrm{RR}}(u)}. \tag{202}$$

The martingale Berry–Esseen bound controls $d_{K(X_n,\mathcal{N}(0,1))}$; the symmetry $-\mathcal{N}(0,1) \stackrel{d}{=} \mathcal{N}(0,1)$ of the standard normal makes the sign of the leading term irrelevant ($d_{K(X_n,\mathcal{N})} = d_{K(-X_n,\mathcal{N})}$).

Smoothing inequality. For random variables X, Y on the same probability space and any $t > 0$,

$$d_K(X+Y, \mathcal{N}(0,1)) \le d_K(X, \mathcal{N}(0,1)) + \mathbb{P}(|Y| > t) + \frac{t}{\sqrt{2\pi}}, \tag{203}$$

where the third term uses the $1/\sqrt{2\pi}$ Lipschitz constant of the standard normal cdf (Bobkov–Götze; see Samsonov et al. 2025, Proposition 12 for an LSA-tailored statement). Bounding $\mathbb{P}(|Y| > t) \le \|\, Y \,\|_{L_p}^p / t^p$ via Markov's inequality and choosing $t = e \,\|\, Y \,\|_{L_p}$ gives the **working form**

$$d_K(X+Y, \mathcal{N}(0,1)) \le d_K(X, \mathcal{N}(0,1)) + \frac{e \,\|\, Y \,\|_{L_p}}{\sqrt{2\pi}} + e^{-p}, \tag{204}$$

in which the trailing tail probability e^{-p} is absorbed into $O(1/n)$ as soon as $p \ge \log n$.

***L_p*-bound on the composite remainder.** The two pieces of $\mathcal{R}_{n,\,\mathrm{stat}}^{\mathrm{RR}}$ contribute additively. The finite-start deterministic transient displayed in Section 4.1 is deliberately excluded: setting $\theta_0 = \theta^*$ would remove only that deterministic term, but not the startup discrepancy between zero-start perturbation variables and the stationary augmented chain.

Lemma 15. Assume the hypotheses and step-size restrictions of the stationary PR-averaged RR misadjustment bound above. Then for every $u \in \mathbb{R}^d$, every $p \geq 2$, and every admissible $q \geq 2$,

$$\| u^\top \mathcal{R}^{\mathrm{RR}}_{n,\,\mathrm{stat}} \|_{L_p} \leq \frac{C_{\mathrm{D2}} \| u \|}{\sqrt{n}} + \| u \| C_{\mathrm{mis}} \Big(\sqrt{n}\, \alpha^2 + \big(1 + d^{1/q}\big) p^{7/2} t_{\mathrm{mix}}^{5/2} \sqrt{n}\, \alpha^{3/2} \log^{3/2}(1/(\alpha a)) \\ + p^{3/2} \sqrt{\alpha} + p^3 (\alpha n)^{-1/2} \log^{1/p}(1/(\alpha a)) + \Phi(p, \alpha)\, n^{-1/2} \Big), \tag{205}$$

with constants

$$C_{\mathrm{D2}} := 3\, t_{\mathrm{mix}} \| \epsilon \|_\infty \left(C_{\mathcal{Q}} + C_2/a^2\right), \tag{206}$$

and C_{mis} the constant in the misadjustment theorem.

Proof. Triangle inequality on $u^\top \mathcal{R}^{\mathrm{RR}}_{n,\,\mathrm{stat}}$:

(a) The deterministic sup-norm bound for $D^{\mathrm{RR}}_{2,n}$ yields $\| u^\top D^{\mathrm{RR}}_{2,n} \|_{L_p} \leq \| u \| \| D^{\mathrm{RR}}_{2,n} \|_\infty \leq C_{\mathrm{D2}} \| u \| / \sqrt{n}$ for every $p \geq 1$.

(b) The misadjustment theorem gives the remaining summands directly. □

Theorem 3. (Stationary $n_0 = 0$ PR-averaged RR Berry–Esseen bound.) Assume **UGE 1**, $\pi\big(\tilde{A}\big) = 0$, $\pi(\epsilon) = 0$, $\| \epsilon \|_\infty < \infty$, $\sigma^2(u) > 0$, $\alpha, 2\alpha \in (0, \alpha_\infty]$, and the stationary augmented-chain convention above. Set $p = \max(2, \lceil \log n \rceil)$ and $q = \max(2p, \lceil \log(e\, d) \rceil, 2)$. If n is large enough that the variance lower-bound condition Eq. 153 holds and $2\alpha \leq \alpha_*(q, t_{\mathrm{mix}})$, then

$$d_K\left(\frac{S^{\mathrm{RR}}_{n,\,\mathrm{stat}}(u)}{\sigma^{\mathrm{RR}}_n(u)}, \mathcal{N}(0,1) \right) \leq \frac{C_{K,1}(u) \log^{3/4} n}{n^{1/4}} + \frac{C_{K,2}(u) \log n}{\sqrt{n}} \\ + \frac{e \| u^\top \mathcal{R}^{\mathrm{RR}}_{n,\,\mathrm{stat}} \|_{L_p}}{\sqrt{2\pi}\, \sigma^{\mathrm{RR}}_n(u)} + \frac{e}{n}, \tag{207}$$

with $\| u^\top \mathcal{R}^{\mathrm{RR}}_{n,\,\mathrm{stat}} \|_{L_p}$ bounded by the preceding lemma, and $C_{K,1}(u), C_{K,2}(u)$ the constants of the martingale Berry–Esseen theorem.

Proof. Apply the smoothing inequality with $X = X_n$ and $Y = Y_n$ from the displayed decomposition above. The first two terms come from the martingale Berry–Esseen bound applied to $-X_n$ (using $d_{K(X_n, \mathcal{N})} = d_{K(-X_n, \mathcal{N})}$ by symmetry of the standard normal). The remainder term is $e \| Y_n \|_{L_p} / \sqrt{2\pi} = e \| u^\top \mathcal{R}^{\mathrm{RR}}_{n,\,\mathrm{stat}} \|_{L_p} / \big(\sqrt{2\pi}\, \sigma^{\mathrm{RR}}_n(u)\big)$. Since $p = \max(2, \lceil \log n \rceil) \geq \log n$, the trailing e^{-p} is at most n^{-1}. □

Corollary 5. Let $\alpha = \alpha_n$ be an admissible step-size sequence and set $p_n := \max(2, \lceil \log n \rceil)$, $q_n := \max(2p_n, \lceil \log(e\, d) \rceil, 2)$, and $\Lambda_n := \log(1/(\alpha_n a))$. Assume the step-size restrictions $2\alpha_n \leq \alpha_\infty$ and $2\alpha_n \leq \alpha_*(q_n, t_{\mathrm{mix}})$, and the variance lower-bound condition Eq. 153. If

$$p_n^3 (n\alpha_n)^{-1/2} \Lambda_n^{1/p_n} \to 0, \quad p_n^{7/2} \sqrt{n}\, \alpha_n^{3/2} \Lambda_n^{3/2} \to 0, \tag{208}$$

then

$$d_K\left(\frac{S^{\mathrm{RR}}_{n,\,\mathrm{stat}}(u)}{\sigma^{\mathrm{RR}}_n(u)}, \mathcal{N}(0,1) \right) \to 0. \tag{209}$$

In particular, for a power scale $\alpha_n = c\, n^{-\gamma}$, up to logarithmic factors, the admissible window is

$$\frac{1}{3} < \gamma < 1. \tag{210}$$

Proof. Apply the stationary $n_0 = 0$ Berry–Esseen bound and the preceding remainder lemma with $p = p_n$, $q = q_n$, and $\alpha = \alpha_n$. The Poisson boundary term is $O\big(n^{-1/2}\big)$. The bias term $\sqrt{n}\alpha_n^2$ is smaller than $\sqrt{n}\alpha_n^{3/2}$ for all sufficiently large n, because the admissibility conditions imply $\alpha_n \to 0$. The terms $p_n^{3/2}\sqrt{\alpha_n}$ and $\Phi(p_n, \alpha_n)n^{-1/2}$ are also negligible under the second admissibility condition. The remaining nontrivial terms are exactly the two quantities displayed in the statement. The martingale Berry–Esseen terms $\log^{3/4}(n)n^{-1/4}$ and $\log(n)n^{-1/2}$ vanish independently of α_n. For $\alpha_n = cn^{-\gamma}$, the two displayed conditions reduce, modulo logarithms, to $n^{-(1-\gamma)/2} \to 0$ and $n^{1/2-3\gamma/2} \to 0$, i.e. $\gamma < 1$ and $\gamma > 1/3$. □

Corollary 6. At the balanced scale $\alpha = c\,n^{-1/2}$ with $c > 0$ such that $\alpha, 2\alpha \in (0, \alpha_\infty]$, with $p = \max(2, \lceil \log n \rceil)$ and $q = \max(2p, \lceil \log(e\,d) \rceil, 2)$ satisfying $2\alpha \le \alpha_*(q, t_{\text{mix}})$, the bound Eq. 207 reduces to

$$d_K\left(\frac{S_{n,\text{ stat}}^{\text{RR}}(u)}{\sigma_n^{\text{RR}}(u)}, \mathcal{N}(0,1)\right) \le \frac{C(u)\,\text{polylog}(n)}{n^{1/4}}, \tag{211}$$

where $C(u)$ depends on $\| u \|$, $\sigma(u)$, C_Q, $\| \overline{A} \|$, $\| \overline{A}^{-1} \|$, κ_Q, C_A, $\| \epsilon \|_\infty$, $\| \Sigma_\epsilon^{(\text{M})} \|$, t_{mix}, a, α_∞, c, and the universal constants of the smoothing, Bolthausen–Fan, and Levin Markov-concentration inequalities.

Proof. The scale $\alpha = c\,n^{-1/2}$ satisfies Eq. 208, but gives the sharper balanced rate. Indeed, $C_{\text{D2}} \| u \| / \sqrt{n} = O\big(n^{-1/2}\big)$; $\sqrt{n}\alpha^2 = c^2\,n^{-1/2}$; $p^{7/2}\sqrt{n}\alpha^{3/2}\log^{3/2}(1/(\alpha a)) = O\big(\text{polylog}(n)n^{-1/4}\big)$; $p^{3/2}\sqrt{\alpha} = O\big(\text{polylog}(n)n^{-1/4}\big)$; $p^3(\alpha n)^{-1/2}\log^{1/p}(1/(\alpha a)) = O\big(\text{polylog}(n)n^{-1/4}\big)$; and $\Phi(p,\alpha)n^{-1/2} = O\big(\text{polylog}(n)n^{-1/2}\big)$. After division by $\sigma_n^{\text{RR}}(u) \ge \sigma(u)/\sqrt{2}$, these terms are dominated by $\text{polylog}(n)n^{-1/4}$. Combining with the martingale Berry–Esseen term proves the claim. □

Corollary 7. Under the hypotheses of the stationary $n_0 = 0$ Berry–Esseen bound above, the same finite-n bound with asymptotic normalisation contains one additional variance-comparison term:

$$\begin{aligned} d_K\left(\frac{S_{n,\text{ stat}}^{\text{RR}}(u)}{\sigma(u)}, \mathcal{N}(0,1)\right) \le{}& \frac{C_{K,1}(u)\log^{3/4} n}{n^{1/4}} + \frac{C_{K,2}(u)\log n}{\sqrt{n}} \\ &+ \frac{e \,\| u^\top \mathcal{R}_{n,\text{ stat}}^{\text{RR}} \|_{L_p}}{\sqrt{2\pi}\,\sigma_n^{\text{RR}}(u)} + \frac{e}{n} + \frac{C \,\| u \|^2}{n\,\alpha\,a\,\sigma^2(u)}. \end{aligned} \tag{212}$$

Consequently, under the balanced-scale hypotheses above,

$$d_K\left(\frac{S_{n,\text{ stat}}^{\text{RR}}(u)}{\sigma(u)}, \mathcal{N}(0,1)\right) \le \frac{C'(u)\,\text{polylog}(n)}{n^{1/4}}. \tag{213}$$

Proof. Set $r := \sigma_n^{\text{RR}}(u)/\sigma(u)$ and write $W := S_{n,\text{ stat}}^{\text{RR}}(u)/\sigma_n^{\text{RR}}(u)$, so $Wr = S_{n,\text{ stat}}^{\text{RR}}(u)/\sigma(u)$. Under the variance lower-bound condition Eq. 153, $r \in \big[1/\sqrt{2}, r_{\max}\big]$ for some finite $r_{\max}$ (from the trivial upper bound $\sigma_n^{2,\text{ RR}}(u) \le C_Q^2 \| \Sigma \| \| u \|^2$). For r in this compact interval,

$$\sup_x | \Phi(x/r) - \Phi(x) | \le C_\Phi\,|r-1|, \quad C_\Phi := \sqrt{2}/\sqrt{\pi e}, \tag{214}$$

because $\sup_x |x\,\varphi(x)| = 1/\sqrt{2\pi e}$. Hence

$$d_K(Wr, \mathcal{N}(0,1)) \le d_K(W, \mathcal{N}(0,1)) + C_\Phi\,|r-1|. \tag{215}$$

The variance comparison of Section 4.5 gives

$$|r-1| = \frac{|\sigma_n^{2,\text{ RR}}(u) - \sigma^2(u)|}{\sigma(u)\,(\sigma_n^{\text{RR}}(u) + \sigma(u))} \le \frac{C_3 \| u \|^2}{n\,\alpha\,a\,\sigma^2(u)}, \tag{216}$$

using $\sigma_n^{\text{RR}}(u) + \sigma(u) \ge \sigma(u)$. This proves the displayed asymptotic-normalisation bound. At $\alpha = cn^{-1/2}$ the final term is $O\big(n^{-1/2}\big)$, hence it is absorbed into the balanced-scale bound of the preceding corollary. □

The bound above is therefore an $n_0 = 0$ stationary augmented-chain statement for $S^{\mathrm{RR}}_{n,\,\mathrm{stat}}(u)$, not the deterministic-start RR average itself. A deterministic-start theorem with logarithmic burn-in requires a separate transfer theorem with $Q^{(\alpha)}_{l,n_0}$ and the corresponding Poisson, variance-comparison, and misadjustment bounds.

5. Burn-in Transfer for Deterministic Starts

The previous chapter proves a stationary $n_0 = 0$ Berry–Esseen bound for $S^{\mathrm{RR}}_{n,\,\mathrm{stat}}(u)$. This chapter transfers that bound to the deterministic-start Richardson–Romberg average after burn-in. Burn-in changes the deterministic PR kernels, so the result is not obtained by simply inserting $n_0 > 0$ into the stationary theorem.

Throughout this chapter, constants with a `"burn"` subscript are independent of n, n_0, m, α, p, and q unless these variables appear as arguments. Dependencies on fixed problem constants and imported Levin/startup constants are absorbed into named constants; powers of a, t_{mix}, p, q, and d are kept explicit when they affect the final rate.

5.1. Target Statistic and Normalization

Let $0 \le n_0 < n$ and set $m := n - n_0$. The burned-in PR average is

$$\overline{\theta}^{(\alpha)}_{n,n_0} := \frac{1}{m} \sum_{k=n_0}^{n-1} \theta^{(\alpha)}_k, \tag{217}$$

and the two-level RR average is

$$\overline{\theta}^{(\mathrm{RR},\alpha)}_{n,n_0} := 2\overline{\theta}^{(\alpha)}_{n,n_0} - \overline{\theta}^{(2\alpha)}_{n,n_0}. \tag{218}$$

The finite-start burned-in vector statistic is

$$\mathcal{T}^{\mathrm{RR}}_{n,n_0} := \sqrt{m}\left(\overline{\theta}^{(\mathrm{RR},\alpha)}_{n,n_0} - \theta^*\right). \tag{219}$$

Its scalar projection in direction $u \in \mathbb{R}^d$ is

$$T^{\mathrm{RR}}_{n,n_0}(u) := u^\top \mathcal{T}^{\mathrm{RR}}_{n,n_0} = \sqrt{m}\, u^\top \left(\overline{\theta}^{(\mathrm{RR},\alpha)}_{n,n_0} - \theta^*\right). \tag{220}$$

The stationary augmented-chain statistic from the previous chapter is used only as a comparison object after the startup transfer.

There are two normalizations. The finite-window normalization is

$$\sigma^{\mathrm{bRR}}_{n,n_0}(u) := \sqrt{\sigma^{2,\,\mathrm{bRR}}_{n,n_0}(u)}, \quad \Xi^{\mathrm{bRR}}_{n,n_0}(u) := \frac{T^{\mathrm{RR}}_{n,n_0}(u)}{\sigma^{\mathrm{bRR}}_{n,n_0}(u)}, \tag{221}$$

where $\sigma^{2,\,\mathrm{bRR}}_{n,n_0}(u)$ is the deterministic variance proxy defined in Eq. 248. The asymptotic normalization is

$$\sigma(u) := \sqrt{u^\top \Sigma_\infty u}, \quad \Xi^{\mathrm{asy,RR}}_{n,n_0}(u) := \frac{T^{\mathrm{RR}}_{n,n_0}(u)}{\sigma(u)}. \tag{222}$$

We first control $\Xi^{\mathrm{bRR}}_{n,n_0}(u)$ and then pass to $\Xi^{\mathrm{asy,RR}}_{n,n_0}(u)$. A final corollary converts the $\sqrt{m}$ statistic to the thesis-facing $\sqrt{n}$ statistic when the burn-in window is logarithmic.

5.2. Burned-in Deterministic Weights

With the $\sqrt{m}$ normalization, the depth-zero linearized term has the weights

$$Q^{(\alpha)}_{l;n_0,n} := \alpha \sum_{k=\max(n_0,l)}^{n-1} B_\alpha^{k-l}, \quad 1 \le l \le n-1, \tag{223}$$

and $Q^{(\alpha)}_{l;n_0,n} = 0$ for $l \ge n$. The RR weight is

$$Q^{\mathrm{RR}}_{l;n_0,n} := 2Q^{(\alpha)}_{l;n_0,n} - Q^{(2\alpha)}_{l;n_0,n}. \tag{224}$$

Thus the leading burned-in sum is

$$W_{n,n_0}^{\mathrm{RR}} := -\frac{1}{\sqrt{m}} \sum_{l=1}^{n-1} Q_{l;n_0,n}^{\mathrm{RR}} \epsilon(Z_l). \tag{225}$$

For $l \geq n_0$ these weights are full-window weights with horizon $n - l$. For $l < n_0$, the lower summation limit is n_0 rather than l, so the weight comparison, Poisson decomposition, and variance proxy must be restated for $Q_{l;n_0,n}^{\mathrm{RR}}$.

5.3. Closed Forms and Burned-in Weight Bounds

Write $B_w := I - w\overline{A}$ for $w \in \{\alpha, 2\alpha\}$ and recall $m = n - n_0$. The burned-in weights have two closed forms. If $l \geq n_0$, then

$$Q_{l;n_0,n}^{(w)} = w \sum_{k=l}^{n-1} B_w^{k-l} = \overline{A}^{-1}\big(I - B_w^{n-l}\big). \tag{226}$$

If $l < n_0$, then

$$Q_{l;n_0,n}^{(w)} = w \sum_{k=n_0}^{n-1} B_w^{k-l} = \overline{A}^{-1}\big(B_w^{n_0-l} - B_w^{n-l}\big). \tag{227}$$

Thus only the post-burn-in weights are close to the asymptotic kernel $\overline{A}^{-1}$. The pre-burn-in weights are instead exponentially small as one moves backward from n_0. Empty sums below are interpreted as zero.

Lemma 16. Assume $\alpha, 2\alpha \in (0, \alpha_\infty]$ and the Lyapunov contraction Eq. 12. Let

$$\rho_\alpha := \sqrt{1 - \alpha a}, \quad C_Q := \kappa_Q^{1/2} \parallel \overline{A}^{-1} \parallel, \quad \tilde{C}_A := \kappa_Q \parallel \overline{A} \parallel. \tag{228}$$

(i) Post-burn-in comparison. If $l \geq n_0$ and $k := n - l$, then

$$\parallel Q_{l;n_0,n}^{\mathrm{RR}} - \overline{A}^{-1} \parallel \leq 3C_Q \rho_\alpha^k. \tag{229}$$

(ii) Pre-burn-in energy. If $1 \leq l < n_0$ and $r := n_0 - l$, then

$$\parallel Q_{l;n_0,n}^{\mathrm{RR}} \parallel \leq 6C_Q \rho_\alpha^r. \tag{230}$$

Consequently, there are constants $C_{\mathrm{burn,E}}$ and $C_{\mathrm{burn,V}}$, depending only on κ_Q, $\parallel \overline{A}^{-1} \parallel$, and $\parallel \overline{A} \parallel$, such that, uniformly in n and n_0,

$$\sum_{l=1}^{n_0-1} \parallel Q_{l;n_0,n}^{\mathrm{RR}} \parallel^2 + \sum_{l=\max(1,n_0)}^{n-1} \parallel Q_{l;n_0,n}^{\mathrm{RR}} - \overline{A}^{-1} \parallel^2 \leq \frac{C_{\mathrm{burn,E}}}{\alpha a}, \tag{231}$$

and

$$\sum_{l=1}^{n-2} \parallel Q_{l+1;n_0,n}^{\mathrm{RR}} - Q_{l;n_0,n}^{\mathrm{RR}} \parallel \leq \frac{C_{\mathrm{burn,V}}}{a^2}. \tag{232}$$

Proof. For $l \geq n_0$, Eq. 226 gives

$$Q_{l;n_0,n}^{\mathrm{RR}} - \overline{A}^{-1} = -\overline{A}^{-1}\big(2B_\alpha^k - B_{2\alpha}^k\big). \tag{233}$$

The same contraction argument as in the full-window estimate gives Eq. 229.

For $l < n_0$, set $r := n_0 - l$. By Eq. 227,

$$Q_{l;n_0,n}^{\mathrm{RR}} = \overline{A}^{-1}\big(2(B_\alpha^r - B_\alpha^{r+m}) - (B_{2\alpha}^r - B_{2\alpha}^{r+m})\big). \tag{234}$$

Taking norms and using $\| B_{2\alpha}^j \|_Q \le \rho_\alpha^j$ gives Eq. 230. Squaring Eq. 229 and Eq. 230 and summing the resulting geometric series gives Eq. 231.

It remains to bound the total variation. In the post-burn-in region $l, l+1 \ge n_0$, the full-window identity gives

$$Q_{l+1;n_0,n}^{\mathrm{RR}} - Q_{l;n_0,n}^{\mathrm{RR}} = -2\alpha(B_\alpha^s - B_{2\alpha}^s), \quad s := n - l - 1. \tag{235}$$

The elementary identity $X^s - Y^s = (X - Y)\sum_{i=1}^s X^{i-1}Y^{s-i}$ with $X = B_\alpha$ and $Y = B_{2\alpha}$ yields, for $s \ge 1$,

$$\| B_\alpha^s - B_{2\alpha}^s \| \le \alpha \tilde{C}_A \, s \, \rho_\alpha^{s-1}. \tag{236}$$

Thus the post-burn-in contribution is bounded by a constant times a^{-2}, as in the full-window proof.

For $l, l+1 < n_0$, put $s := n_0 - l - 1$. The pre-burn-in closed form gives

$$Q_{l+1;n_0,n}^{\mathrm{RR}} - Q_{l;n_0,n}^{\mathrm{RR}} = 2\alpha(B_\alpha^s - B_{2\alpha}^s - B_\alpha^{s+m} + B_{2\alpha}^{s+m}). \tag{237}$$

Here $s \ge 1$. Applying Eq. 236 to the powers s and $s+m$ and summing over $s \ge 1$ again gives a constant times a^{-2}. If the boundary $l = n_0 - 1$ exists, then

$$Q_{n_0;n_0,n}^{\mathrm{RR}} - Q_{n_0-1;n_0,n}^{\mathrm{RR}} = 2\alpha(B_{2\alpha}^m - B_\alpha^m), \tag{238}$$

and Eq. 236 with $s = m$ bounds this term by the same a^{-2} scale. Combining the pre-burn-in, boundary, and post-burn-in contributions proves Eq. 232. □

5.4. Deterministic Transient After Burn-in

For a generic step size $w \in \{\alpha, 2\alpha\}$ define the deterministic transient

$$D_{\mathrm{tr},n,n_0}^{(w)} := \frac{1}{\sqrt{m}} \sum_{k=n_0}^{n-1} B_w^k(\theta_0 - \theta^*), \quad B_w := I - w\overline{A}. \tag{239}$$

The Richardson–Romberg transient entering Eq. 220 is

$$D_{\mathrm{tr},n,n_0}^{\mathrm{RR}}(u) := u^\top \left(2D_{\mathrm{tr},n,n_0}^{(\alpha)} - D_{\mathrm{tr},n,n_0}^{(2\alpha)}\right). \tag{240}$$

Lemma 17. Assume $\alpha, 2\alpha \in (0, \alpha_\infty]$ and the Lyapunov contraction Eq. 12. Then, for each $w \in \{\alpha, 2\alpha\}$,

$$\| D_{\mathrm{tr},n,n_0}^{(w)} \| \le \frac{2\sqrt{\kappa_Q}}{wa\sqrt{m}}(1 - wa)^{n_0/2} \| \theta_0 - \theta^* \|. \tag{241}$$

Consequently,

$$| D_{\mathrm{tr},n,n_0}^{\mathrm{RR}}(u) | \le \frac{5\sqrt{\kappa_Q} \| u \| \| \theta_0 - \theta^* \|}{\alpha a \sqrt{m}}(1 - \alpha a)^{n_0/2}. \tag{242}$$

Proof. Let $r_w := \sqrt{1 - wa}$. By Eq. 12 and equivalence of the Euclidean and Q-norms,

$$\| B_w^k(\theta_0 - \theta^*) \| \le \sqrt{\kappa_Q} r_w^k \| \theta_0 - \theta^* \|. \tag{243}$$

Therefore

$$\| D_{\mathrm{tr},n,n_0}^{(w)} \| \le \frac{\sqrt{\kappa_Q} \| \theta_0 - \theta^* \|}{\sqrt{m}} \sum_{k=n_0}^{n-1} r_w^k \le \frac{\sqrt{\kappa_Q} \| \theta_0 - \theta^* \|}{\sqrt{m}} \frac{r_w^{n_0}}{1 - r_w}. \tag{244}$$

Since $1 - \sqrt{1 - x} \ge x/2$ for $0 \le x \le 1$, the last display gives Eq. 241. Applying this estimate at $w = \alpha$ and $w = 2\alpha$, using $1 - 2\alpha a \le 1 - \alpha a$, and taking the triangle inequality in Eq. 240 gives

$$| \ D^{\mathrm{RR}}_{\mathrm{tr},n,n_0}(u) \ | \le \| \ u \ \| \ \Big(2 \ \| \ D^{(\alpha)}_{\mathrm{tr},n,n_0} \ \| + \| \ D^{(2\alpha)}_{\mathrm{tr},n,n_0} \ \|\Big) \le \frac{5\sqrt{\kappa_Q} \ \| \ u \ \| \ \| \ \theta_0 - \theta^* \ \|}{\alpha a \sqrt{m}} (1-\alpha a)^{n_0/2}. \tag{245}$$

□

Corollary 8. (Logarithmic burn-in removes the deterministic transient.) If, for some $\beta > 0$,

$$n_0 \ge \frac{2\beta}{\alpha a} \log n, \tag{246}$$

then

$$| \ D^{\mathrm{RR}}_{\mathrm{tr},n,n_0}(u) \ | \le \frac{5\sqrt{\kappa_Q} \ \| \ u \ \| \ \| \ \theta_0 - \theta^* \ \|}{\alpha a \sqrt{m}} n^{-\beta}. \tag{247}$$

In particular, at the balanced scale $\alpha = cn^{-1/2}$ and $m \ge n/2$, this is $O\big(n^{-\beta}\big)$. Taking $\beta = 1$ makes the deterministic transient negligible relative to the stationary $n^{-1/4}$ Berry–Esseen rate.

Proof. The elementary inequality $(1-\alpha a)^{n_0/2} \le \exp(-\alpha a n_0/2)$ and Eq. 246 imply $(1-\alpha a)^{n_0/2} \le n^{-\beta}$. Substitute this into Eq. 242. If $\alpha = cn^{-1/2}$ and $m \ge n/2$, then $\big(\alpha\sqrt{m}\big)^{-1} \le \sqrt{2}/c$, so the displayed $O\big(n^{-\beta}\big)$ bound follows. □

5.5. Burned-in Variance Proxy

For the stochastic depth-zero term, write $Q^{\mathrm{bRR}}_l := Q^{\mathrm{RR}}_{l;n_0,n}$ and $\Sigma := \Sigma^{(\mathrm{M})}_{\epsilon}$. We take $n \ge 2$ in this subsection. The martingale part produced by the Poisson decomposition below starts at $l = 2$, so the deterministic variance proxy is

$$\Sigma^{\mathrm{bRR}}_{n,n_0} := \frac{1}{m} \sum_{l=2}^{n-1} Q^{\mathrm{bRR}}_l \, \Sigma \left(Q^{\mathrm{bRR}}_l\right)^\top, \quad \sigma^{2,\,\mathrm{bRR}}_{n,n_0}(u) := u^\top \Sigma^{\mathrm{bRR}}_{n,n_0} u. \tag{248}$$

Lemma 18. Assume the conditions of Lemma 16 and $\| \ \Sigma \ \| < \infty$. There exists a constant $C_{\mathrm{burn},3}$, depending only on κ_Q, $\| \ \overline{A}^{-1} \ \|$, $\| \ \overline{A} \ \|$, and $\| \ \Sigma \ \|$, such that

$$\| \ \Sigma^{\mathrm{bRR}}_{n,n_0} - \Sigma_\infty \ \| \le \frac{C_{\mathrm{burn},3}}{m \, \alpha a}. \tag{249}$$

Consequently,

$$| \ \sigma^{2,\,\mathrm{bRR}}_{n,n_0}(u) - \sigma^2(u) \ | \le \frac{C_{\mathrm{burn},3} \ \| \ u \ \|^2}{m \, \alpha a}. \tag{250}$$

Proof. For post-burn-in indices put $\Delta_l := Q^{\mathrm{bRR}}_l - \overline{A}^{-1}$. Then

$$Q^{\mathrm{bRR}}_l \Sigma \big(Q^{\mathrm{bRR}}_l\big)^\top - \Sigma_\infty = \Delta_l \Sigma \overline{A}^{-\top} + \overline{A}^{-1} \Sigma \Delta_l^\top + \Delta_l \Sigma \Delta_l^\top. \tag{251}$$

By Eq. 229,

$$\sum_{l \ge \max(2,n_0)} \| \ \Delta_l \ \| \le \frac{C}{\alpha a}, \quad \sum_{l \ge \max(2,n_0)} \| \ \Delta_l \ \|^2 \le \frac{C}{\alpha a}. \tag{252}$$

For pre-burn-in indices there is no comparison with $\overline{A}^{-1}$; instead Eq. 231 gives

$$\sum_{2 \le l < n_0} \| \ Q^{\mathrm{bRR}}_l \ \|^2 \le \frac{C}{\alpha a}. \tag{253}$$

Replacing all post-burn-in weights by $\overline{A}^{-1}$ gives at most m copies of Σ_∞; the martingale index set starts at $l = 2$, so the possible finite-index mismatch is bounded by two copies of Σ_∞. Combining these three estimates and dividing by m yields Eq. 249, after absorbing the harmless $2\,\|\Sigma_\infty\frac{\|}{m}$ term into $\frac{C_{\mathrm{burn},3}}{m\alpha a}$ using $\alpha a \le 1$. The scalar bound follows from $|u^\top H u| \le \|H\|\,\|u\|^2$. □

Assume $\sigma^2(u) > 0$ and impose the burned-in variance lower-bound condition

$$m\,\alpha\,a \ge \frac{2C_{\mathrm{burn},3}\,\|\,u\,\|^2}{\sigma^2(u)}. \tag{254}$$

Then Eq. 250 gives $\sigma^{2,\,\mathrm{bRR}}_{n,n_0}(u) \ge \sigma^2(u)/2$, hence $\sigma^{\mathrm{bRR}}_{n,n_0}(u) \ge \sigma(u)/\sqrt{2}$. At the balanced scale $\alpha = c\,n^{-1/2}$ with $m \ge n/2$, this condition holds for all sufficiently large n depending on u, c, and the problem constants.

5.6. Burned-in Poisson Martingale Approximation

Let $\hat{\epsilon} := \sum_{j=0}^{\infty} Q^j \epsilon$ be the Poisson solution, so

$$\hat{\epsilon} - Q\hat{\epsilon} = \epsilon, \quad \|\,\hat{\epsilon}\,\|_\infty \le 3\,t_{\mathrm{mix}}\,\|\,\epsilon\,\|_\infty. \tag{255}$$

The stationary Poisson identity applies because the coefficients Q_l^{bRR} are deterministic.

Lemma 19. Assume **UGE 1** and $\pi(\epsilon) = 0$. Define, for $2 \le l \le n-1$,

$$\Delta M_l^{\mathrm{bRR}} := Q_l^{\mathrm{bRR}}\,(\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1})), \tag{256}$$

and set $M^{\mathrm{bRR}}_{n,n_0} := \sum_{l=2}^{n-1} \Delta M_l^{\mathrm{bRR}}$. Then $\left\{\Delta M_l^{\mathrm{bRR}}\right\}_{l=2}^{n-1}$ is a sequence of $\mathcal{F}_l$-martingale differences and

$$W^{\mathrm{RR}}_{n,n_0} = -\frac{1}{\sqrt{m}} M^{\mathrm{bRR}}_{n,n_0} + D^{\mathrm{bRR}}_{2,n,n_0}, \tag{257}$$

where

$$D^{\mathrm{bRR}}_{2,n,n_0} := -\frac{1}{\sqrt{m}}\left[Q_1^{\mathrm{bRR}}\,\hat{\epsilon}(Z_1) + \sum_{l=1}^{n-2}\big(Q_{l+1}^{\mathrm{bRR}} - Q_l^{\mathrm{bRR}}\big)\,Q\hat{\epsilon}(Z_l)\right]. \tag{258}$$

Moreover, with $C_{\mathrm{burn,Q}} := \|\,\overline{A}^{-1}\,\| + 6C_Q$,

$$\|\,D^{\mathrm{bRR}}_{2,n,n_0}\,\|_\infty \le \frac{3\,t_{\mathrm{mix}}\,\|\,\epsilon\,\|_\infty}{\sqrt{m}}\left(C_{\mathrm{burn,Q}} + \frac{C_{\mathrm{burn,V}}}{a^2}\right). \tag{259}$$

Proof. The martingale-difference property follows from the Markov property: $\mathbb{E}[\hat{\epsilon}(Z_l) \mid \mathcal{F}_{l-1}] = Q\hat{\epsilon}(Z_{l-1})$.

Substitute the Poisson equation in Eq. 225. The $l = 1$ term is kept as a left boundary term, and for $l \ge 2$ we add and subtract $Q\hat{\epsilon}(Z_{l-1})$. Abel summation of the telescope gives exactly Eq. 257. The right boundary vanishes because $Q^{\mathrm{RR}}_{n-1;n_0,n} = 2\alpha I - 2\alpha I = 0$.

For the sup-norm bound, use $\|\,Q\hat{\epsilon}\,\|_\infty \le \|\,\hat{\epsilon}\,\|_\infty$, the uniform weight bound $\|\,Q_l^{\mathrm{bRR}}\,\| \le C_{\mathrm{burn,Q}}$ from Eq. 229 and Eq. 230, and the total-variation estimate Eq. 232. □

5.7. Burned-in Predictable Variance Comparison

For the martingale in Lemma 19 define

$$\mathcal{V}_\epsilon(z) := Q(\hat{\epsilon}\hat{\epsilon}^\top)(z) - (Q\hat{\epsilon})(z)\,(Q\hat{\epsilon})(z)^\top. \tag{260}$$

This is the same matrix-valued conditional covariance function as in the stationary chapter; it is not the vector noise ϵ. As before, $\pi(\mathcal{V}_\epsilon) = \Sigma$ and

$$\|\,\mathcal{V}_\epsilon\,\|_\infty \le 18\,t_{\mathrm{mix}}^2\,\|\,\epsilon\,\|_\infty^2. \tag{261}$$

The predictable quadratic variation is

$$\langle M^{\mathrm{bRR}}\rangle_{n,n_0} := \sum_{l=2}^{n-1} \mathbb{E}\Big[\Delta M_l^{\mathrm{bRR}} \big(\Delta M_l^{\mathrm{bRR}}\big)^\top \mid \mathcal{F}_{l-1}\Big] = \sum_{l=2}^{n-1} Q_l^{\mathrm{bRR}}\, \mathcal{V}_\epsilon(Z_{l-1})\, \big(Q_l^{\mathrm{bRR}}\big)^\top. \tag{262}$$

Lemma 20. Assume **UGE 1**, $\pi(\epsilon) = 0$, and $\|\, \epsilon \,\|_\infty < \infty$. There exists a universal constant $C_4 > 0$ such that, for every $u \in \mathbb{R}^d$, every $p \geq 2$, every initial distribution ξ, every $n \geq 2$, and every $0 \leq n_0 < n$,

$$\mathbb{E}_\xi^{1/p}\Big[\mid u^\top \langle M^{\mathrm{bRR}}\rangle_{n,n_0} u - m\, \sigma^{2,\,\mathrm{bRR}}_{n,n_0}(u) \mid^p\Big] \leq C_4\, C^2_{\mathrm{burn,Q}} \,\|\, u \,\|^2 \,\|\, \epsilon \,\|^2_\infty\, t_{\mathrm{mix}}^{5/2} \sqrt{p\, n}. \tag{263}$$

Proof. Set

$$h_{l(z)} := u^\top Q_l^{\mathrm{bRR}}\, \mathcal{V}_\epsilon(z) \big(Q_l^{\mathrm{bRR}}\big)^\top u, \quad g_{l(z)} := h_{l(z)} - \pi(h_l). \tag{264}$$

By Eq. 261 and $\|\, Q_l^{\mathrm{bRR}} \,\| \leq C_{\mathrm{burn,Q}}$,

$$\|\, g_l \,\|_\infty \leq 36\, C^2_{\mathrm{burn,Q}} \,\|\, u \,\|^2\, t^2_{\mathrm{mix}} \,\|\, \epsilon \,\|^2_\infty. \tag{265}$$

Moreover, Eq. 262 and Eq. 248 imply

$$u^\top \langle M^{\mathrm{bRR}}\rangle_{n,n_0} u - m\, \sigma^{2,\,\mathrm{bRR}}_{n,n_0}(u) = \sum_{l=2}^{n-1} g_{l(Z_{l-1})}. \tag{266}$$

The Markov concentration lemma used in the stationary chapter, applied to the time-inhomogeneous centered functions g_l, gives the displayed $\sqrt{pn}$ bound. □

Corollary 9. Under the assumptions of Lemma 20,

$$\mathbb{E}_\xi^{1/p}\Big[\mid u^\top \langle M^{\mathrm{bRR}}\rangle_{n,n_0} u - m\, \sigma^2(u) \mid^p\Big] \leq C_4\, C^2_{\mathrm{burn,Q}} \,\|\, u \,\|^2 \,\|\, \epsilon \,\|^2_\infty\, t_{\mathrm{mix}}^{5/2} \sqrt{p\, n} + \frac{C_{\mathrm{burn,3}} \,\|\, u \,\|^2}{\alpha a}. \tag{267}$$

Proof. Use the triangle inequality and Eq. 250:

$$m\, |\sigma^{2,\,\mathrm{bRR}}_{n,n_0}(u) - \sigma^2(u)| \leq \frac{C_{\mathrm{burn,3}} \,\|\, u \,\|^2}{\alpha a}. \tag{268}$$

□

5.8. Startup Transfer for Augmented-Chain Remainders

After the deterministic transient is removed, a startup discrepancy remains: the finite-start perturbation variables $J^{(\ell,w)}$ and $H^{(2,w)}$ start from zero, whereas the stationary theorem uses the invariant law of the augmented chain. This discrepancy is summed over the post-burn-in window.

For $w \in \{\alpha, 2\alpha\}$ write

$$Y_k^{(w)} := \Big(Z_{k+1}, J_k^{(0,w)}, J_k^{(1,w)}, J_k^{(2,w)}\Big). \tag{269}$$

For $y = (z, j_0, j_1, j_2)$ and $y' = (z', j_0', j_1', j_2')$ define the depth-two augmented-chain cost used by Levin et al. (2025, Appendix B.2, Eq. (49)):

$$\begin{aligned} c^{(w)}_{J,2}(y, y') := {} & \|j_0 - j_0'\| + \|j_1 - j_1'\| + \|j_2 - j_2'\| \\ & + \big(\|j_0\| + \|j_0'\| + \|j_1\| + \|j_1'\| + \|j_2\| + \|j_2'\| + \sqrt{wa}\, \|\epsilon\|_\infty\big)\, 1_{z \neq z'}. \end{aligned} \tag{270}$$

Levin et al. (2025, Appendix B.2, Proposition 5, with constants defined in their Eq. (55)) prove the Wasserstein contraction for this cost. The componentwise estimates in its proof imply that, under their Proposition-5 step-size restriction, two copies of $Y_k^{(w)}$ started from deterministic states y, y' can be coupled so that, for $\ell = 0, 1, 2$,

$$\| J_k^{(\ell,w)}(y) - J_k^{(\ell,w)}(y') \|_{L_p} \le C_W \, p^{7/2} \, t_{\mathrm{mix}}^{5/2} \log^{3/2}(1/(wa)) \exp(-wak/(12p)) \, c_{J,2}^{(w)}(y,y'). \tag{271}$$

Their Corollary 4 gives the corresponding invariant law $\Pi_{J,2,w}$ for $Y_k^{(w)}$.

The contraction above does not include $H^{(2,w)}$. Levin et al. (2025, Appendix D.1, Proposition 9) prove the required one-trajectory moment bound for $H^{(2,w)}$, using the representation

$$H_k^{(2,w)} = -w \sum_{l=1}^{k} \Gamma_{l+1:k}^{(w)} \tilde{A}(Z_l) J_{l-1}^{(2,w)}. \tag{272}$$

For startup transfer, however, one must compare two copies of this display. The following lemma records the resulting full-state contraction. It is the only place where the $H^{(2,w)}$ coordinate is added to the Levin Proposition-5 coupling.

Let $\alpha_{\mathrm{st}}(p)$ denote the step-size ceiling under which the argument below is valid. It is the minimum of the Proposition-5 restriction for $Y_k^{(w)}$ and the product-stability restriction used in Levin Proposition 9 with moments up to order $2p$. For $w \in \{\alpha, 2\alpha\}$ write

$$R_{k,\,\mathrm{fin}}^{(w)} := J_{k,\,\mathrm{fin}}^{(1,w)} + J_{k,\,\mathrm{fin}}^{(2,w)} + H_{k,\,\mathrm{fin}}^{(2,w)} \tag{273}$$

for the finite-start depth-two remainder, and define $R_{k,\,\mathrm{aug}}^{(w)}$ analogously for a copy initialized from the invariant law of the augmented chain $\left(Z_{k+1}, J_k^{(0,w)}, J_k^{(1,w)}, J_k^{(2,w)}, H_k^{(2,w)}\right)$.

Lemma 21. (Full-state startup contraction for the depth-two augmented remainder.) Assume **UGE 1**, $\pi(\epsilon) = 0$, $\|\epsilon\|_\infty < \infty$, the step-size restrictions of the Levin depth-two moment bounds, and $2\alpha \le \alpha_{\mathrm{st}}(p)$. There exist constants $c_{\mathrm{st}} > 0$ and $C_{\mathrm{st}} < \infty$, depending only on the problem constants and on the constants in Eq. 271 and Levin Proposition 9, such that for every $p \ge 2$, every admissible $q \ge 2$, every initial distribution ξ of the base chain with finite-start perturbation coordinates initialized at zero, every $w \in \{\alpha, 2\alpha\}$, and every $k \ge 0$, the finite-start and stationary augmented remainders can be coupled so that

$$\| R_{k,\,\mathrm{fin}}^{(w)} - R_{k,\,\mathrm{aug}}^{(w)} \|_{L_p} \le A_{\mathrm{st}}(p,q,w) \exp(-c_{\mathrm{st}} wak/p), \tag{274}$$

where

$$A_{\mathrm{st}}(p,q,w) := C_{\mathrm{st}}\big(1 + d^{1/q}\big) p^7 t_{\mathrm{mix}}^5 \sqrt{w/a} \log^{3/2}(1/(wa)). \tag{275}$$

Proof. Work on the exact-coupling probability space used in Levin Appendix B.2. Let T be the coupling time of the two base chains. By UGE, $\mathbb{P}(T > r) \le C\rho^r$, and Proposition 5 gives, for $\ell \in \{0,1,2\}$,

$$\| J_{k,\,\mathrm{fin}}^{(\ell,w)} - J_{k,\,\mathrm{aug}}^{(\ell,w)} \|_{L_p} \le C p^{7/2} t_{\mathrm{mix}}^{5/2} \log^{3/2}(1/(wa)) e^{-cwak/p} \mathbb{E}^{1/p}\left[c_{J,2}^{(w)}\big(Y_0^{\mathrm{fin}}, Y_0^{\mathrm{aug}}\big)^p\right]. \tag{276}$$

The finite-start coordinates are zero. The invariant copy satisfies the finite-past limits of the elementary $J^{(0,w)}$, $J^{(1,w)}$ estimates and of Levin Propositions 8 and 9; in particular

$$\mathbb{E}^{1/p}\left[c_{J,2}^{(w)}\big(Y_0^{\mathrm{fin}}, Y_0^{\mathrm{aug}}\big)^p\right] \le C\big(1 + d^{1/q}\big) p^{7/2} t_{\mathrm{mix}}^{5/2} \sqrt{w/a} \log^{3/2}(1/(wa)). \tag{277}$$

Hence the $J^{(1,w)}$ and $J^{(2,w)}$ parts of $R_k^{(w)}$ satisfy Eq. 274 after enlarging C_{st}.

It remains to control $H^{(2,w)}$. On the event $T \le k$, the base chains are identical after time T, and subtracting Eq. 272 gives

$$\Delta H_k^{(2,w)} = \Gamma_{T+1:k}^{(w)} \Delta H_T^{(2,w)} - w \sum_{l=T+1}^{k} \Gamma_{l+1:k}^{(w)} \tilde{A}(Z_l) \Delta J_{l-1}^{(2,w)}, \tag{278}$$

where Δ denotes finite-start minus stationary. On $T > k$ we simply use Holder inequality, the tail bound for T, and the one-trajectory Levin Proposition 9 moment bound for the two $H^{(2,w)}$ copies. This gives

$$\| \Delta H_k^{(2,w)} 1_{T>k} \|_{L_p} \leq C\big(1 + d^{1/q}\big)p^{7/2}t_{\text{mix}}^{5/2}w^{3/2} \log^{3/2}(1/(wa))e^{-ck/p}. \tag{279}$$

Since $wa \leq 1$, this is bounded by the right-hand side of Eq. 274.

For the first term in Eq. 278, combine the random-product estimate used in Levin Proposition 9 with the same one-trajectory moment bound:

$$\| \Gamma_{T+1:k}^{(w)} \Delta H_T^{(2,w)} 1_{T\leq k} \|_{L_p} \leq C\big(1 + d^{1/q}\big)p^{7/2}t_{\text{mix}}^{5/2}w^{3/2} \log^{3/2}(1/(wa))e^{-cwak/p}. \tag{280}$$

Here the exponential moment of T is finite because $w \leq \alpha_{\text{st}}(p)$.

For the convolution term, use the same product estimate and the $J^{(2,w)}$ contraction Eq. 276:

$$\begin{aligned} &w \sum_{l=1}^{k} \| \Gamma_{l+1:k}^{(w)} \tilde{A}(Z_l) \Delta J_{l-1}^{(2,w)} \|_{L_p} \\ &\leq CwA_{\text{st}}(p, q, w) \sum_{l=1}^{k} e^{-cwa(k-l)} e^{-cwal/p} \\ &\leq CA_{\text{st}}(p, q, w)e^{-cwak/p}. \end{aligned} \tag{281}$$

The last line absorbs the geometric convolution into the constant after decreasing c; fixed powers of a^{-1} are included in C_{st}. Combining Eq. 279, Eq. 280, and Eq. 281 gives the same startup bound for $H^{(2,w)}$. Adding the already controlled $J^{(1,w)}$ and $J^{(2,w)}$ pieces proves Eq. 274. □

Define the RR startup discrepancy accumulated over the burned-in window by

$$\mathcal{U}_{n,n_0}^{\text{start,RR}} := \frac{1}{\sqrt{m}} \sum_{k=n_0}^{n-1} \Big[2\Big(R_{k,\,\text{fin}}^{(\alpha)} - R_{k,\,\text{aug}}^{(\alpha)}\Big) - \Big(R_{k,\,\text{fin}}^{(2\alpha)} - R_{k,\,\text{aug}}^{(2\alpha)}\Big)\Big]. \tag{282}$$

Lemma 22. (Accumulated startup transfer.) Under Lemma 21, if $\alpha, 2\alpha \in (0, \alpha_\infty]$ and $\alpha a \leq 1/4$, then for every $p \geq 2$ and admissible $q \geq 2$,

$$\| \mathcal{U}_{n,n_0}^{\text{start,RR}} \|_{L_p} \leq \frac{C_{\text{start,RR}} p\, A_{\text{st}}(p, q, \alpha)}{\alpha a \sqrt{m}} \exp(-c_{\text{st}} \alpha a n_0/p). \tag{283}$$

Proof. Apply Eq. 274 at $w = \alpha$ and $w = 2\alpha$, then use the triangle inequality:

$$\| \mathcal{U}_{n,n_0}^{\text{start,RR}} \|_{L_p} \leq \frac{1}{\sqrt{m}} \sum_{k=n_0}^{n-1} \big(2A_{\text{st}}(p, q, \alpha)e^{-c_{\text{st}}\alpha ak/p} + A_{\text{st}}(p, q, 2\alpha)e^{-2c_{\text{st}}\alpha ak/p}\big). \tag{284}$$

Since $\alpha a \leq 1/4$, $A_{\text{st}}(p, q, 2\alpha) \leq CA_{\text{st}}(p, q, \alpha)$ and $1 - \exp(-c_{\text{st}}\alpha a/p) \geq C^{-1}\alpha a/p$. Extending the geometric sum to infinity gives Eq. 283. □

Corollary 10. (Logarithmic burn-in removes the startup discrepancy.) If, for some $\beta > 0$,

$$n_0 \geq \frac{\beta p}{c_{\mathrm{st}} \alpha a} \log n, \tag{285}$$

then

$$\| \mathcal{U}_{n,n_0}^{\mathrm{start,RR}} \|_{L_p} \leq \frac{C_{\mathrm{start,RR}} p\, A_{\mathrm{st}}(p,q,\alpha)}{\alpha a \sqrt{m}} n^{-\beta}. \tag{286}$$

At the balanced scale $\alpha = cn^{-1/2}$, with $m \geq n/2$ and p, q logarithmic in n, this is $\mathrm{polylog}(n)\, n^{-1/4-\beta}$. Thus the Berry–Esseen choice $p \asymp \log n$ requires n_0 of order $(\alpha a)^{-1} \log^2 n$ under this L_p contraction.

Proof. Substitute Eq. 285 into Eq. 283. For the balanced scale, $A_{\mathrm{st}}(p,q,\alpha) = \mathrm{polylog}(n)\, \alpha^{1/2}$, hence $A_{\mathrm{st}}(p,q,\alpha)/(\alpha\sqrt{m}) = O\big(\mathrm{polylog}(n) n^{-1/4}\big)$. □

5.9. Burned-in Depth-Two Misadjustment Bound

Define the finite-start burned-in RR misadjustment by

$$R_{n,n_0,\,\mathrm{fin}}^{\mathrm{mis,RR}} := \frac{1}{\sqrt{m}} \sum_{k=n_0}^{n-1} \Big(2R_{k,\,\mathrm{fin}}^{(\alpha)} - R_{k,\,\mathrm{fin}}^{(2\alpha)}\Big). \tag{287}$$

For comparison, let

$$R_{m,\,\mathrm{aug}}^{\mathrm{mis,RR}} := \frac{1}{\sqrt{m}} \sum_{j=0}^{m-1} \Big(2R_{j,\,\mathrm{aug}}^{(\alpha)} - R_{j,\,\mathrm{aug}}^{(2\alpha)}\Big) \tag{288}$$

be the stationary augmented-chain depth-two misadjustment over a window of length m. By stationarity, the same distribution is obtained if the sum in Eq. 288 is taken over $j = n_0, ..., n-1$.

Theorem 4. (Burned-in PR-averaged RR misadjustment bound.) Assume **UGE 1**, $\pi(\epsilon) = 0$, $\| \epsilon \|_\infty < \infty$, $\alpha, 2\alpha \in (0, \alpha_\infty]$, $\alpha a \leq 1/4$, and the step-size restrictions of the Levin depth-two and startup-contraction bounds. Set

$$\Phi(p,\alpha) := p^{3/2}\, t_{\mathrm{mix}}^{1/2}/a + p^{1/2}\, t_{\mathrm{mix}}^{3/2} \sqrt{\alpha/a}. \tag{289}$$

There exists a constant $C_{\mathrm{burn,mis}}$ depending only on the stationary misadjustment constants, the startup-contraction constants, and the problem constants such that, for every $p \geq 2$, every $q \geq 2$ satisfying $p \leq q/2$, every $2\alpha \leq \alpha_*(q, t_{\mathrm{mix}})$ and $2\alpha \leq \alpha_{\mathrm{st}}(p)$, and every $m \geq 2$,

$$\begin{aligned}
\| R_{n,n_0,\,\mathrm{fin}}^{\mathrm{mis,RR}} \|_{L_p} \leq\; & C_{\mathrm{burn,mis}} \sqrt{m}\, \alpha^2 \\
& + C_{\mathrm{burn,mis}} \big(1 + d^{1/q}\big) p^{7/2} t_{\mathrm{mix}}^{5/2} \sqrt{m}\, \alpha^{3/2} \log^{3/2}(1/(\alpha a)) \\
& + C_{\mathrm{burn,mis}} p^{3/2} \sqrt{\alpha} \\
& + C_{\mathrm{burn,mis}} p^3 (\alpha m)^{-1/2} \log^{1/p}(1/(\alpha a)) \\
& + C_{\mathrm{burn,mis}} \Phi(p,\alpha)\, m^{-1/2} \\
& + \frac{C_{\mathrm{burn,mis}} p\, A_{\mathrm{st}}(p,q,\alpha)}{\alpha a \sqrt{m}} \exp(-c_{\mathrm{st}} \alpha a n_0 / p).
\end{aligned} \tag{290}$$

Proof. Couple the finite-start and stationary augmented-chain remainders as in Lemma 21, and define the stationary window on the same indices by

$$\tilde{R}^{\mathrm{mis,RR}}_{n,n_0,\,\mathrm{aug}} := \frac{1}{\sqrt{m}} \sum_{k=n_0}^{n-1} \Big(2R^{(\alpha)}_{k,\,\mathrm{aug}} - R^{(2\alpha)}_{k,\,\mathrm{aug}} \Big). \tag{291}$$

Then, with the notation of Eq. 282,

$$R^{\mathrm{mis,RR}}_{n,n_0,\,\mathrm{fin}} = \tilde{R}^{\mathrm{mis,RR}}_{n,n_0,\,\mathrm{aug}} + \mathcal{U}^{\mathrm{start,RR}}_{n,n_0} \tag{292}$$

under this coupling. Therefore

$$\| \, R^{\mathrm{mis,RR}}_{n,n_0,\,\mathrm{fin}} \, \|_{L_p} \le \| \, \tilde{R}^{\mathrm{mis,RR}}_{n,n_0,\,\mathrm{aug}} \, \|_{L_p} + \| \, \mathcal{U}^{\mathrm{start,RR}}_{n,n_0} \, \|_{L_p}. \tag{293}$$

By stationarity, $\| \, \tilde{R}^{\mathrm{mis,RR}}_{n,n_0,\,\mathrm{aug}} \, \|_{L_p} = \| \, R^{\mathrm{mis,RR}}_{m,\,\mathrm{aug}} \, \|_{L_p}$, and the latter is exactly the stationary augmented-chain misadjustment bound Theorem 2 with n replaced by m. The second term is Lemma 22. Combining the two estimates gives Eq. 290. □

Corollary 11. (Balanced-scale burned-in misadjustment rate.) Assume the hypotheses of Theorem 4. Let $\alpha = c\, n^{-1/2}$, $m \ge n/2$, $p = \max(2, \lceil \log n \rceil)$, and $q = \max(2p, \lceil \log(e\, d) \rceil, 2)$. If, for some fixed $\beta > 0$,

$$n_0 \ge \frac{\beta p}{c_{\mathrm{st}} \alpha a} \log n, \tag{294}$$

and n is large enough that $2\alpha \le \alpha_*(q, t_{\mathrm{mix}})$ and $2\alpha \le \alpha_{\mathrm{st}}(p)$, then

$$\| \, R^{\mathrm{mis,RR}}_{n,n_0,\,\mathrm{fin}} \, \|_{L_p} \le C_{\mathrm{burn,mis}} \, \mathrm{polylog}(n) \, n^{-1/4}. \tag{295}$$

Proof. The stationary part of Eq. 290 is the bound of Corollary 4 with n replaced by m, and $m \ge n/2$ keeps the rate unchanged. The startup term is controlled by Corollary 10 and is $\mathrm{polylog}(n)\, n^{-1/4-\beta}$ at this scale. □

5.10. Burned-in Martingale Berry–Esseen

The depth-zero martingale in Lemma 19 has deterministic coefficients, but its normalization is based on the effective sample size $m = n - n_0$. We assume $m \ge n/2$, the logarithmic burn-in regime, so the inhomogeneous concentration term of order $\sqrt{pn}$ can be written on the m scale.

Fix $u \in \mathbb{R}^d$ and put $X^{\mathrm{bRR}}_l := u^\top \Delta M^{\mathrm{bRR}}_l$ for $2 \le l \le n-1$. Then

$$|X^{\mathrm{bRR}}_l| \le \kappa_{\mathrm{burn}}(u), \quad \kappa_{\mathrm{burn}}(u) := 6\, t_{\mathrm{mix}}\, C_{\mathrm{burn,Q}} \, \| \, \epsilon \, \|_\infty \, \| \, u \, \|. \tag{296}$$

Indeed, this is the same bounded-increment estimate as in the stationary chapter, with $C_{\mathrm{burn,Q}}$ replacing the full-window weight bound. Set

$$s^2_{n,n_0}(u) := m\, \sigma^{2,\,\mathrm{bRR}}_{n,n_0}(u). \tag{297}$$

Under Eq. 254, $s^2_{n,n_0}(u) \ge m\sigma^2(u)/2$. Also, $m \ge n/2$ and the uniform bound $\|Q^{\mathrm{bRR}}_l\| \le C_{\mathrm{burn,Q}}$ imply the deterministic upper bound

$$s^2_{n,n_0}(u) \le 2m\, C^2_{\mathrm{burn,Q}} \, \| \, \Sigma^{(\mathrm{M})}_\epsilon \, \| \, \| \, u \, \|^2. \tag{298}$$

Theorem 5. (Burned-in martingale Berry–Esseen bound.) Assume **UGE 1**, $\pi(\epsilon) = 0$, $\| \, \epsilon \, \|_\infty < \infty$, $\sigma^2(u) > 0$, $\alpha, 2\alpha \in (0, \alpha_\infty]$, $m \ge n/2$, and the burned-in variance lower-bound condition Eq. 254. There exist constants $C_{\mathrm{bK},1}(u), C_{\mathrm{bK},2}(u) > 0$, depending only on $\|u\|$, $\sigma(u)$, $C_{\mathrm{burn,Q}}$, t_{mix}, $\|\epsilon\|_\infty$, $\|\Sigma^{(\mathrm{M})}_\epsilon\|$, and the universal Bolthausen–Fan constants, such that for every $n \ge 3$,

$$d_K\left(\frac{u^\top M^{\mathrm{bRR}}_{n,n_0}}{\sqrt{m}\, \sigma^{\mathrm{bRR}}_{n,n_0}(u)}, \mathcal{N}(0,1) \right) \le \frac{C_{\mathrm{bK},1}(u) \log^{3/4} n}{m^{1/4}} + \frac{C_{\mathrm{bK},2}(u) \log n}{\sqrt{m}}. \tag{299}$$

Proof. Apply the Bolthausen–Fan inequality used in the stationary chapter to the martingale differences X_l^{bRR}. The bounded-increment input is Eq. 296, and the target variance is $s^2_{n,n_0}(u)$. The first and third Bolthausen–Fan terms are bounded exactly as before, with n replaced by m in the denominator and the harmless factor $n/m \le 2$ absorbed into the constants. For the conditional-variance term use Lemma 20:

$$\mathbb{E}^{1/p}\Big[\mid u^\top \langle M^{\mathrm{bRR}}\rangle_{n,n_0} u - s^2_{n,n_0}(u) \mid^p\Big] \le C\sqrt{p\,n}. \tag{300}$$

Together with $m \ge n/2$, Eq. 254, and Eq. 298, this gives

$$\text{Term II} \le C(u)\, p^{(3p+1)/(2(2p+1))}\, m^{-p/(2(2p+1))}. \tag{301}$$

Taking $p = \lceil \log n \rceil$ gives the displayed $\log^{3/4}(n)m^{-1/4}$ bound. The classical Bolthausen and Lindeberg terms give the second displayed term. □

5.11. Finite-Window Smoothing Assembly

The finite-start scalar statistic decomposes as

$$T^{\mathrm{RR}}_{n,n_0}(u) = -\frac{u^\top M^{\mathrm{bRR}}_{n,n_0}}{\sqrt{m}} + \mathcal{R}^{\mathrm{bRR}}_{n,n_0,\mathrm{fin}}(u), \tag{302}$$

where the composite non-Gaussian remainder is

$$\mathcal{R}^{\mathrm{bRR}}_{n,n_0,\mathrm{fin}}(u) := D^{\mathrm{RR}}_{\mathrm{tr},n,n_0}(u) + u^\top D^{\mathrm{bRR}}_{2,n,n_0} + u^\top R^{\mathrm{mis,RR}}_{n,n_0,\mathrm{fin}}. \tag{303}$$

The terms are controlled by Lemma 17, Lemma 19, and Theorem 4, respectively.

Let $\mathcal{B}_{\mathrm{mis}}(m, n_0, p, q, \alpha)$ denote the right-hand side of Eq. 290.

Lemma 23. Assume the hypotheses of Lemma 17, Lemma 19, and Theorem 4. Then, for every $p \ge 2$ and admissible $q \ge 2$,

$$\begin{aligned}\| \mathcal{R}^{\mathrm{bRR}}_{n,n_0,\mathrm{fin}}(u) \|_{L_p} \le{}& \frac{5\sqrt{\kappa_Q}\,\| u \|\,\| \theta_0 - \theta^* \|}{\alpha a\sqrt{m}}(1-\alpha a)^{n_0/2} \\ &+ \frac{C_{\mathrm{burn,D2}}\,\| u \|}{\sqrt{m}} + \| u \|\,\mathcal{B}_{\mathrm{mis}}(m, n_0, p, q, \alpha),\end{aligned} \tag{304}$$

where

$$C_{\mathrm{burn,D2}} := 3\,t_{\mathrm{mix}}\,\| \epsilon \|_\infty \left(C_{\mathrm{burn,Q}} + \frac{C_{\mathrm{burn,V}}}{a^2}\right). \tag{305}$$

Proof. Apply the triangle inequality in Eq. 303. The deterministic part is Eq. 242. The Poisson boundary term is bounded by

$$\| u^\top D^{\mathrm{bRR}}_{2,n,n_0} \|_{L_p} \le \|u\|\,\|D^{\mathrm{bRR}}_{2,n,n_0}\|_\infty \le \frac{C_{\mathrm{burn,D2}}\,\|u\|}{\sqrt{m}} \tag{306}$$

using Eq. 259. The last term is exactly Eq. 290. □

Dividing Eq. 302 by the finite-window normalization gives

$$\Xi^{\mathrm{bRR}}_{n,n_0}(u) = X^{\mathrm{bRR}}_{n,n_0} + Y^{\mathrm{bRR}}_{n,n_0}, \tag{307}$$

where

$$X^{\mathrm{bRR}}_{n,n_0} := -\frac{u^\top M^{\mathrm{bRR}}_{n,n_0}}{\sqrt{m}\,\sigma^{\mathrm{bRR}}_{n,n_0}(u)}, \quad Y^{\mathrm{bRR}}_{n,n_0} := \frac{\mathcal{R}^{\mathrm{bRR}}_{n,n_0,\mathrm{fin}}(u)}{\sigma^{\mathrm{bRR}}_{n,n_0}(u)}. \tag{308}$$

Theorem 6. (Finite-window burned-in PR-averaged RR Berry–Esseen bound.) Assume the hypotheses of Theorem 5 and Theorem 4. Let $p = \max(2, \lceil \log n \rceil)$ and $q = \max(2p, \lceil \log(e\,d) \rceil, 2)$. If $2\alpha \le \alpha_*(q, t_{\mathrm{mix}})$ and $2\alpha \le \alpha_{\mathrm{st}}(p)$, then

$$\begin{aligned} d_K\Big(\Xi^{\mathrm{bRR}}_{n,n_0}(u), \mathcal{N}(0,1)\Big) \le \frac{C_{\mathrm{bK},1}(u)\log^{3/4} n}{m^{1/4}} + \frac{C_{\mathrm{bK},2}(u)\log n}{\sqrt{m}} \\ + \frac{e\,\|\, \mathcal{R}^{\mathrm{bRR}}_{n,n_0,\mathrm{fin}}(u)\,\|_{L_p}}{\sqrt{2\pi}\,\sigma^{\mathrm{bRR}}_{n,n_0}(u)} + \frac{e}{n}, \end{aligned} \tag{309}$$

with the composite remainder bounded by Lemma 23.

Proof. Apply the smoothing inequality Eq. 204 to the split Eq. 307. The martingale Berry–Esseen term is Theorem 5; the minus sign in X^{bRR}_{n,n_0} is irrelevant because the standard normal law is symmetric. Since $p = \max(2, \lceil \log n \rceil)$, the smoothing tail e^{-p} is at most e/n. The L_p norm of the perturbation is

$$\|Y^{\mathrm{bRR}}_{n,n_0}\|_{L_p} = \frac{\|\mathcal{R}^{\mathrm{bRR}}_{n,n_0,\mathrm{fin}}(u)\|_{L_p}}{\sigma^{\mathrm{bRR}}_{n,n_0}(u)}, \tag{310}$$

which gives Eq. 309. □

5.12. Balanced Burn-in Berry–Esseen Bound

The finite-window bound uses $\sigma^{\mathrm{bRR}}_{n,n_0}(u)$. The final inference statement uses $\sigma(u)$, via the following burned-in analogue of Corollary 7.

Lemma 24. Assume $\sigma^2(u) > 0$ and the burned-in variance lower-bound condition Eq. 254. Put

$$r_{n,n_0}(u) := \frac{\sigma^{\mathrm{bRR}}_{n,n_0}(u)}{\sigma(u)}. \tag{311}$$

Then

$$\frac{1}{\sqrt{2}} \le r_{n,n_0}(u) \le \sqrt{3/2}, \tag{312}$$

and, for any real random variable W,

$$d_K\Big(r_{n,n_0}(u)W, \mathcal{N}(0,1)\Big) \le d_K(W, \mathcal{N}(0,1)) + \frac{C_{\mathrm{norm}} C_{\mathrm{burn},3}\,\|u\|^2}{m\,\alpha\,a\,\sigma^2(u)}, \tag{313}$$

where C_{norm} is a universal constant.

Proof. The variance lower-bound condition and Eq. 250 give

$$|\sigma^{2,\mathrm{bRR}}_{n,n_0}(u) - \sigma^2(u)| \le \sigma^2(u)/2. \tag{314}$$

Hence $r_{n,n_0}(u) \in \Big[1/\sqrt{2}, \sqrt{3/2}\Big]$. For r in this compact interval,

$$\sup_x |\ \Phi(x/r) - \Phi(x)\ | \le C_{\mathrm{norm}}\,|r-1|, \tag{315}$$

because $\sup_x |x\varphi(x)| < \infty$. Therefore

$$d_K(rW, \mathcal{N}(0,1)) \le d_K(W, \mathcal{N}(0,1)) + C_{\mathrm{norm}}\,|r-1|. \tag{316}$$

Finally,

$$|r_{n,n_0}(u) - 1| = \frac{|\sigma^{2,\,\mathrm{bRR}}_{n,n_0}(u) - \sigma^2(u)|}{\sigma(u)\left(\sigma^{\mathrm{bRR}}_{n,n_0}(u) + \sigma(u)\right)} \le \frac{C_{\mathrm{burn},3}\ \|u\|^2}{m\,\alpha\,a\,\sigma^2(u)}, \tag{317}$$

using Eq. 250 and $\sigma^{\mathrm{bRR}}_{n,n_0}(u) + \sigma(u) \ge \sigma(u)$. □

Theorem 7. (Balanced-scale burned-in PR-averaged RR Berry–Esseen bound.) Assume Assumptions 1–3 from Section 1.4. Assume also the Lyapunov contraction Eq. 12 for the two step sizes used below, the Levin depth-two stationary moment and misadjustment bounds used in Theorem 2, and the non-degeneracy condition $\sigma^2(u) > 0$. Fix $c > 0$ and set

$$\alpha := c\,n^{-1/2}, \quad m := n - n_0, \quad p := \max(2, \lceil \log n \rceil), \quad q := \max(2p, \lceil \log(e\,d) \rceil, 2). \tag{318}$$

Put

$$\alpha_{\mathrm{adm}}(p,q) := \min\Big(\alpha_\infty, \frac{1}{2a}, \alpha_*(q, t_{\mathrm{mix}}), \alpha_{\mathrm{st}}(p)\Big). \tag{319}$$

Here $\alpha_*(q, t_{\mathrm{mix}})$ collects the Levin depth-two moment admissibility restrictions, while $\alpha_{\mathrm{st}}(p)$ is the full-state startup threshold in Lemma 21. Suppose $n \ge 3$ is such that

$$m \ge n/2, \quad 2\alpha \le \alpha_{\mathrm{adm}}(p,q), \quad m\,\alpha\,a \ge \frac{2C_{\mathrm{burn},3}\ \|\ u\ \|^2}{\sigma^2(u)}, \tag{320}$$

and the burn-in satisfies the two explicit logarithmic conditions

$$n_0 \ge \frac{2}{\alpha a}\log n, \quad n_0 \ge \frac{p}{c_{\mathrm{st}}\alpha a}\log n. \tag{321}$$

Then there exists a finite constant $C_{\mathrm{burn,final}}(u, c, \theta_0)$, depending only on u, c, $\|\theta_0 - \theta^*\|$, and the problem and universal constants in the preceding bounds, such that

$$d_K\Big(\Xi^{\mathrm{bRR}}_{n,n_0}(u), \mathcal{N}(0,1)\Big) \le \frac{C_{\mathrm{burn,final}}(u,c,\theta_0)\,\mathrm{polylog}(n)}{n^{1/4}}, \tag{322}$$

and

$$d_K\Big(\Xi^{\mathrm{asy,RR}}_{n,n_0}(u), \mathcal{N}(0,1)\Big) \le \frac{C_{\mathrm{burn,final}}(u,c,\theta_0)\,\mathrm{polylog}(n)}{n^{1/4}}. \tag{323}$$

Proof. Apply the finite-window assembly theorem Theorem 6. Since $m \ge n/2$, the martingale terms satisfy

$$\frac{\log^{3/4} n}{m^{1/4}} + \frac{\log n}{\sqrt{m}} \le C\,\frac{\mathrm{polylog}(n)}{n^{1/4}}. \tag{324}$$

It remains to bound the composite remainder in Lemma 23. The first condition in Eq. 321 is Eq. 246 with $\beta = 1$; by Corollary 8 and $\alpha = cn^{-1/2}$, $m \ge n/2$, the deterministic transient is $O(n^{-1})$. The Poisson Abel remainder is $O\big(m^{-1/2}\big)$. The second condition in Eq. 321 is the startup condition Eq. 285 with $\beta = 1$, so Corollary 11 gives

$$\|\ R^{\mathrm{mis,RR}}_{n,n_0,\,\mathrm{fin}}\ \|_{L_p} \le C\,\mathrm{polylog}(n)\,n^{-1/4}. \tag{325}$$

Together with $\sigma^{\mathrm{bRR}}_{n,n_0}(u) \ge \sigma(u)/\sqrt{2}$, this makes the smoothing remainder in Eq. 309 of order $\mathrm{polylog}(n)n^{-1/4}$. The smoothing tail e/n is lower order, so Eq. 322 follows.

For the asymptotic normalization, write

$$\Xi^{\mathrm{asy,RR}}_{n,n_0}(u) = r_{n,n_0}(u)\,\Xi^{\mathrm{bRR}}_{n,n_0}(u). \tag{326}$$

By Lemma 24, the additional cost is at most

$$\frac{C\,\|u\|^2}{m\,\alpha\,a\,\sigma^2(u)} = O\big(n^{-1/2}\big), \tag{327}$$

because $m \geq n/2$ and $\alpha = cn^{-1/2}$. This is absorbed into the balanced finite-window rate, proving Eq. 323. □

Condition Eq. 320 collects the finite-n admissibility requirements. The inequality $2\alpha \leq \alpha_{\mathrm{adm}}(p,q)$ enforces the Lyapunov small-step ceiling, the Levin depth-two admissibility threshold, and the full-state startup contraction Lemma 21. Since $\alpha = cn^{-1/2}$ and $m \geq n/2$, the elementary step-size constraints and Eq. 254 hold automatically for all sufficiently large n. The remaining non-elementary large-n requirement is

$$2cn^{-1/2} \leq \min(\alpha_*(q,t_{\mathrm{mix}}), \alpha_{\mathrm{st}}(p)), \tag{328}$$

with p, q as in the theorem. Under this Levin/startup admissibility condition, the large-n reading of the theorem keeps only $m \geq n/2$ and the two burn-in lower bounds in Eq. 321.

Corollary 12. ($\sqrt{n}$-normalization for the burned-in RR statistic.) Under the assumptions of Theorem 7, define the thesis-facing scalar statistic

$$T^{\mathrm{RR,n}}_{n,n_0}(u) := \sqrt{n}\,u^\top\Big(\overline{\theta}^{(\mathrm{RR},\alpha)}_{n,n_0} - \theta^*\Big) = \sqrt{n/m}\,T^{\mathrm{RR}}_{n,n_0}(u), \tag{329}$$

and its asymptotic normalization

$$\Xi^{\mathrm{n,RR}}_{n,n_0}(u) := \frac{T^{\mathrm{RR,n}}_{n,n_0}(u)}{\sigma(u)}. \tag{330}$$

If, in addition, the burn-in window has the logarithmic upper bound

$$n_0 \leq C_0\,(\alpha a)^{-1}\log^2 n \tag{331}$$

for some finite C_0, then there exists a finite constant $C_{\mathrm{burn,n}}(u,c,\theta_0,C_0)$ such that

$$d_K\Big(\Xi^{\mathrm{n,RR}}_{n,n_0}(u), \mathcal{N}(0,1)\Big) \leq \frac{C_{\mathrm{burn,n}}(u,c,\theta_0,C_0)\,\mathrm{polylog}(n)}{n^{1/4}}. \tag{332}$$

Proof. Put $s_{n,n_0} := \sqrt{n/m}$. Since Eq. 320 gives $m \geq n/2$,

$$0 \leq s_{n,n_0} - 1 = \frac{n_0}{m\,\big(s_{n,n_0}+1\big)} \leq \frac{2n_0}{n}. \tag{333}$$

For every real random variable W and every $s \in \big[1, \sqrt{2}\big]$,

$$d_K(sW, \mathcal{N}(0,1)) \leq d_K(W, \mathcal{N}(0,1)) + C\,|s-1|, \tag{334}$$

by the same scaling argument as in Lemma 24. Applying this with $W = \Xi^{\mathrm{asy,RR}}_{n,n_0}(u)$ and using Theorem 7 gives the already proved $\mathrm{polylog}(n)n^{-1/4}$ term. The upper burn-in bound Eq. 331 and $\alpha = cn^{-1/2}$ give

$$|s_{n,n_0} - 1| \leq \frac{2C_0}{ca}\,\frac{\log^2 n}{n^{1/2}}, \tag{335}$$

which is lower order and is absorbed into the same balanced-scale rate. □

The burn-in lower bounds in Eq. 321 amount to $n_0 \geq C(\alpha a)^{-1}\log^2 n$ when $p \asymp \log n$. This logarithmic square comes from the current L_p startup contraction.